

REF (S) ·

- ## NOTES
1. USE TWO LAG SCREWS AND WASHERS FOR EACH FOUR-WAY STRAIN BAND.
 2. ≠ USE ONLY WHEN SPECIFIED.
 3. Δ ANCHOR TYPE SHACKLE AND 2 1/4" LINK NOT REQUIRED WHEN TYPE B POLE TOP BAND, DWG NO. 41 6035, IS USED.
 4. * WHEN AN ADDITIONAL DISC IS USED ON ANGLE STRUCTURES, AT RAILROAD OR MAIN HIGHWAY CROSSINGS DIMENSION SHALL BE INCREASED ONE FOOT.
 5. Ø GREATER THAN 22-INCH LENGTH MAY BE REQUIRED FOR EXTRA LARGE POLES.

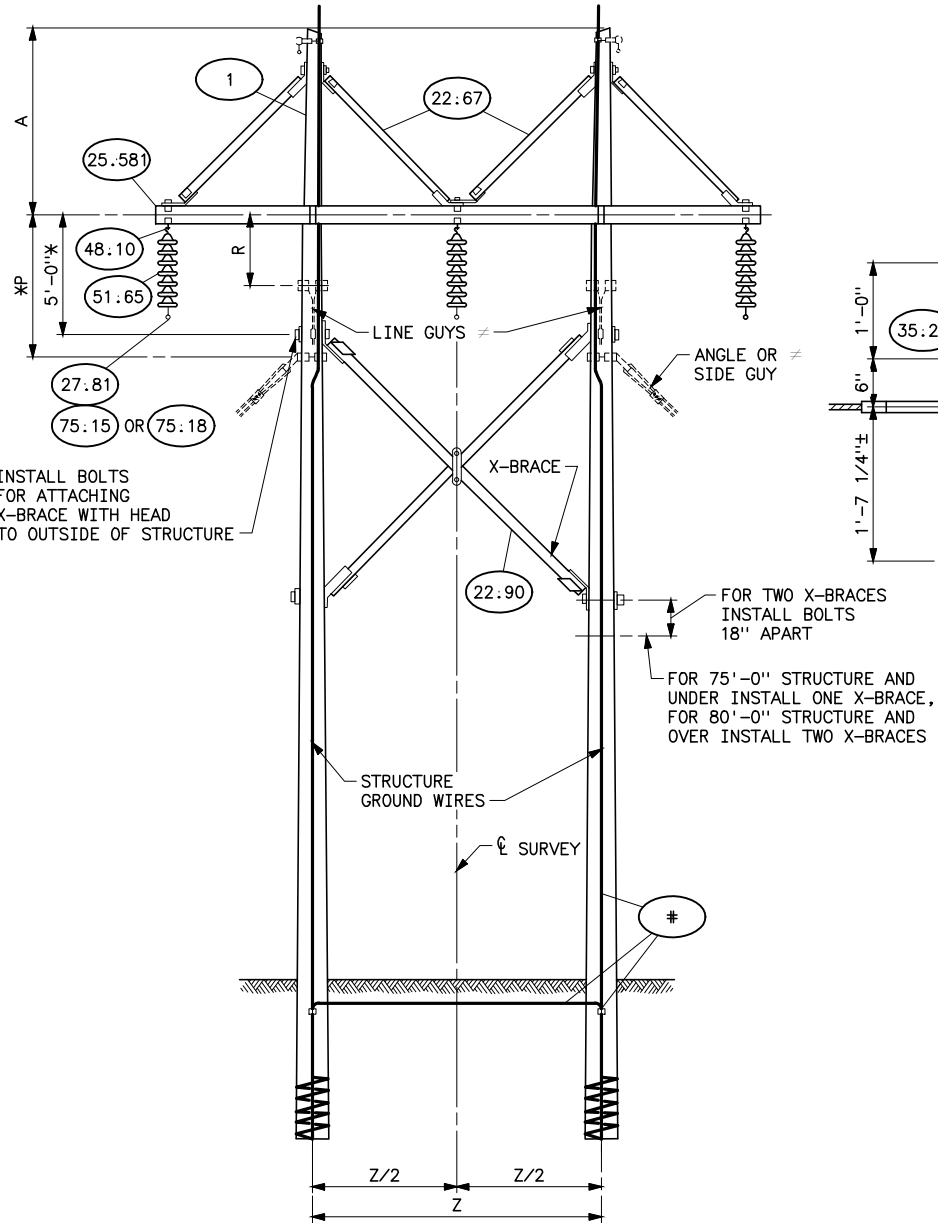
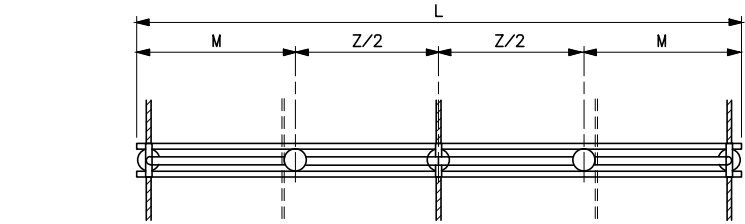
REFERENCE DRAWINGS			
GROUNDING DETAILS	-----	41	1012
DRILLING AND GAINING DETAILS WITH OGW	-----	41	6024
X-BRACE DETAIL	-----	41	6029
H-FRAME ANGLE IRON DETAILS	-----	41	6041
GUY ARRANGEMENT-ASSEMBLY-H-FRAME	-----	41	6052

INSULATOR MATERIAL						
PART NO.	QUANTITY			DESCRIPTION	DWG NO. 41	
	138KV		161KV			
27.81	3		3	CLAMP, SUSPENSION WITH SOCKET FITTING (FOR CONDUCTOR)	1000	
48.10	3		3	HOOK, SUSPENSION	1003	
51.65	24	3 STRINGS OF 8	30	3 STRINGS OF 10	INSULATOR, SUSPENSION TYPE, 10" DIA. 5 3/4" SPACING	1000

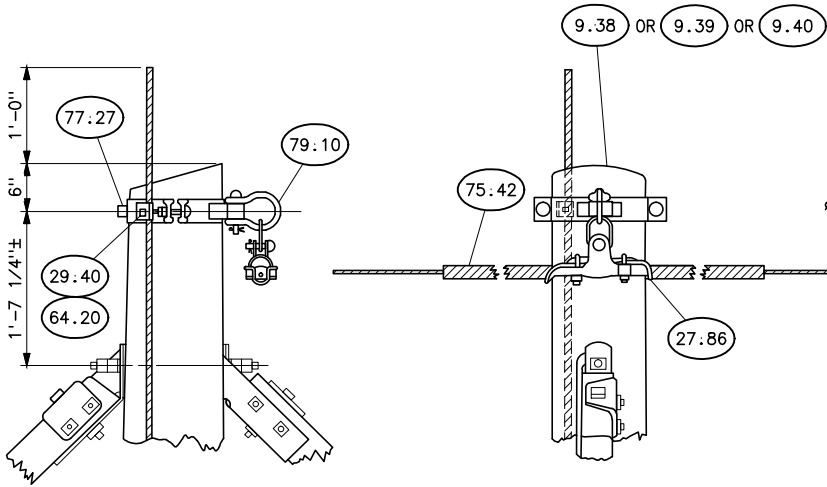
OVERHEAD GROUND WIRE MATERIAL					
PART NO.	QUANTITY			DESCRIPTION	DWG NO. 41
	SUSPENSION	DEAD END	TENSION		
9.20	—	2	2	BAND, FOUR-WAY STRAIN	6030
9.43, 9.44 OR 9.45	2	—	—	BAND, POLE TOP	6033, 6036, 6037
27.86	2Δ	—	—	CLAMP, SUSPENSION, WITH 2 1/4" LINK	
29.40	2	—	—	CLIP, GROUND, WITH 3/8"x1 1/2" CARRIAGE BOLT	6033, 6036, 6037
29.50	—	2	2	CLIP, GROUND, WITH 3/4"x2 1/2" CONE HEAD BOLT	6030
35.20	—	2	4	DEAD END, COMPRESSION TYPE O.G.W.	
58.11 OR 58.13	—	2	4	LINK, CONNECTING, PR. FOR EYE-TYPE DEAD END, USE 58.11 FOR CLEVIS-TYPE DEAD END, USE 58.13	6030
64.20	2	—	—	LOCKNUT FOR 3/8" BOLT	
64.23	2	2	2	LOCKNUT FOR 3/4" BOLT	
77.27	2	4	4	LAG SCREW, 1/2"x4", FETTER DRIVE THREAD	
20.41	2	—	—	BOLT, "U", 5/8"x3" LONG, WITH 4 HEX NUTS	
91.22	—	4	4	WASHER, ROUND FLAT, 1 3/8" O.D., 9/16" HOLE	
64.22	4	—	—	LOCKNUT FOR 5/8" BOLT WITH HEX NUTS	
91.12	4	—	—	LOCKWASHER, SPRING FOR 5/8" BOLT	
91.13	2	—	—	LOCK WASHER, SPRING FOR 3/4" BOLT	
19.71	2	—	—	BOLT MACHINE, 3/4"x12"	
C	1	—	—	ANGLE IRON	6041

B	1-27-14 A7-DDD	REDRAWN.
	SUPERSEDES DWG NO. 40-D-5683 UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO	
TRANSMISSION LINE STANDARDS TYPE HSB 138KV AND 161KV SUSPENSION STRUCTURE		
DESIGNED <u>BUREC</u> _____ APPROVED <u>BUREC</u> _____		
CAE	JUNE 30, 1981	41
		6007

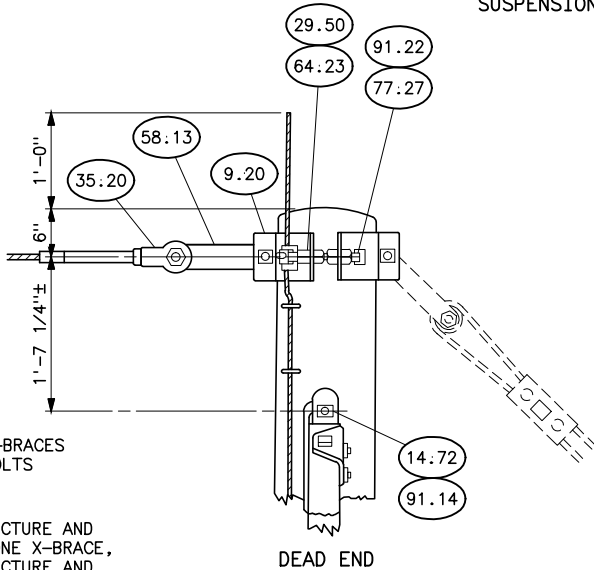
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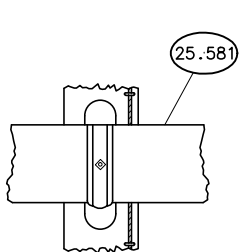
VOLTAGE	A		L	Z	Z/2	M	PX	R
	WITH OGW	WITHOUT OGW						
69 & 115KV	8'-2"	8'-2"	25'-0"	12'-0"	6'-0"	6'-6"	6'-0"	3'-0"



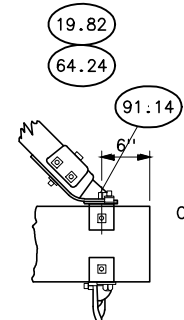
SUSPENSION



DETAIL OF OVERHEAD GROUND WIRE ATTACHMENT



DETAIL OF CROSSARM ATTACHMENT



DETAIL OF PHASE FITTING ASSEMBLY

INSTALL MACHINE BOLT PART NO. 19.32, WITH HEAD DOWN

OMIT ON CENTER PHASE FITTING ASSEMBLY

STRUCTURE MATERIAL						
PART NO.	QUANTITY				DESCRIPTION	DWG NO. 41
	WITH OVERHEAD GROUND WIRES		W/OUT OVERHEAD GROUND WIRES			
	69KV	115KV	69KV	115KV		
1	2	2	2	2	POLE	
25.581	2	2	2	2	CROSSARM, WOOD, 2 5/8"x9 1/2"x25'-0"	6023
14.72	2	2	2	2	BOLT, DOUBLE ARMING, 7/8"x14" WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS	
14.75	2	2	2	2	BOLT, DOUBLE ARMING, 7/8"x20" WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS	
19.32	4	4	4	4	BOLT, MACHINE, 1/2"x12"	
19.45	12	12	12	12	BOLT, MACHINE, 5/8"x5"	
19.82	3	3	3	3	BOLT, MACHINE, 7/8"x3"	
22.67	4	4	4	4	BRACE, KNEE, SPACING 5'-8", DROP 5'-8", 3 3/8"x4 3/8" SECTION	
40.52	3	3	3	3	FITTING, PHASE	6038
64.21	4	4	4	4	LOCKNUT FOR 1/2" BOLT	
64.22	12	12	12	12	LOCKNUT FOR 5/8" BOLT	
64.24	3	3	3	3	LOCKNUT FOR 7/8" BOLT	
68.30	4	4	4	4	PLATE, FLAT GAIN	6038
68.82	4	4	4	4	PLATE, RIBBED TIE	6038
79.50	3	3	3	3	SHACKLE, TWISTED	6038
91.11	4	4	4	4	LOCK WASHER, SPRING, FOR 1/2" BOLT	
91.12	12	12	12	12	LOCK WASHER, SPRING, FOR 5/8" BOLT	
91.14	11	11	11	11	LOCK WASHER, SPRING, FOR 7/8" BOLT	
91.60	4	4	4	4	WASHER, SQUARE FLAT 2"x2"x1/8", 9/16" HOLE	
91.62	12	12	12	12	WASHER, SQUARE FLAT 2 1/4"x2 1/4"x3/16", 11/16" HOLE	
22.90	1 OR 2	1 OR 2	1 OR 2	1 OR 2	X-BRACE, COMPLETE WITH FITTINGS, (3 3/8"x4 3/8")	6029
#	PART NUMBERS AND DESCRIPTION SHOWN ON 41 1012					

INSULATOR MATERIAL

PART NO.	69KV	115KV	DESCRIPTION	DWG NO. 41
27.81	3	3	CLAMP, SUSPENSION, WITH SOCKET FITTING (FOR CONDUCTOR)	1000
48.10	3	3	HOOK, SUSPENSION	1003
51.65	12	21	INSULATOR, SUSPENSION TYPE, 10" DIAMETER, 5 3/4" SPACING	1000
75.15 OR 75.18	3	3	ROD, ARMOR, PREFORMED, SINGLE SUPPORT (FOR ACSR ONLY)	1000

OVERHEAD GROUND WIRE MATERIAL

PART NO.	SUSPENSION	DEADEND	TENSION	DESCRIPTION	DWG NO. 41
9.20	-	2	2	BAND, FOUR-WAY STRAIN	6030
9.38, 9.39 OR 9.40	2	-	-	BAND, POLE TOP	6035, 6036, 6037
29.40	2	-	-	CLIP, GROUND WITH 3/8"x1 1/2" CARRIAGE BOLT	
27.86	2	-	-	CLAMP, SUSPENSION, WITH 2 1/4" LINK	
29.50	-	2	2	CLIP, GROUND WITH 3/4"x2 1/2" CONE HEAD BOLT	6030
35.20	-	2	4	DEAD END, COMPRESSION TYPE, FOR OVERHEAD GROUND WIRE	
58.13	-	2	4	LINK, CONNECTING, PAIR	6030
64.23	-	2	2	LOCKNUT FOR 3/4" BOLT	
77.27	2	4	4	LAG SCREW, 1/2"x4", FETTER DRIVE THREAD	
79.10	2	-	-	SHACKLE, ANCHOR TYPE	6036, 6037
91.22	-	4	4	WASHER, ROUND FLAT, 1-3/8" OD, 9/16" HOLE	
64.20	2	-	-	LOCKNUT FOR 3/8" BOLT	
75.42	2 SETS	-	-	ROD, ARMOR, PREFORMED	

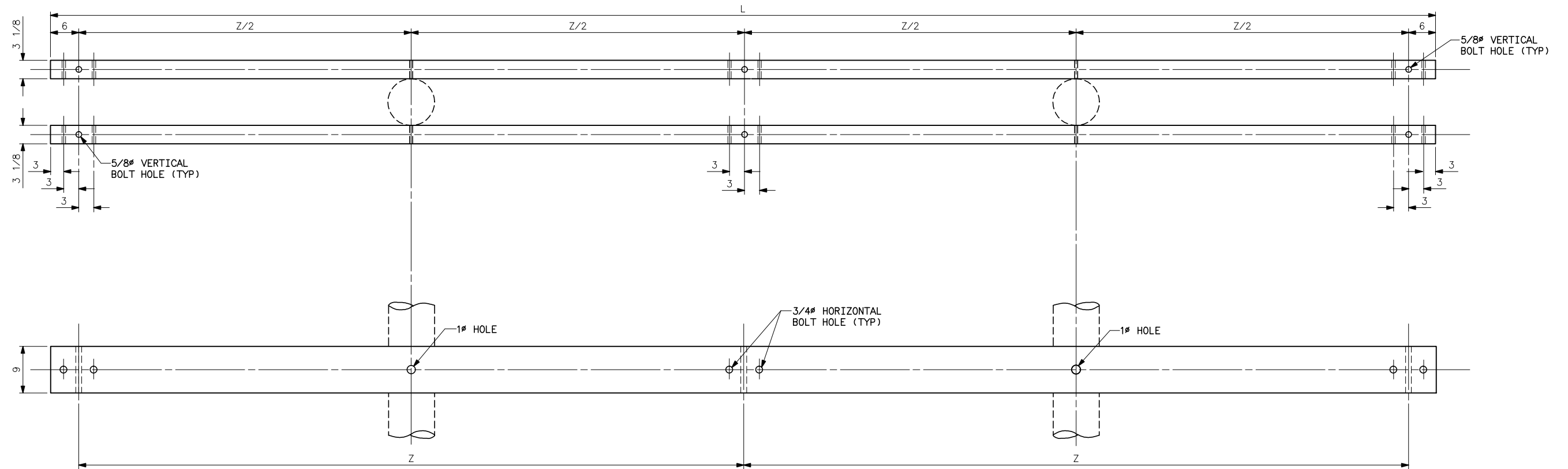
REFERENCE DRAWINGS

GROUNDING DETAILS-----41 1012
DRILLING AND GAINING DETAILS WITH OGW -
30° PROTECTIVE ANGLE-----41 6021
GUY ARRANGEMENT-ASSEMBLY-----
H-FRAME-----41 6051 AND 41 6052
DRILLING AND GAINING DETAILS (WITHOUT OGW)-----41 6066
DRILLING AND GAINING DETAILS WITH OGW -
39° PROTECTIVE ANGLE-----41 6068
GUY ARRANGEMENT-WITH GUY STUB AND
OVERHEAD GROUND WIRE-----41 6086

NOTES

1. USE TWO LAG SCREWS AND WASHERS FOR EACH FOUR-WAY STRAIN BAND.
2. ≠ USE ONLY WHEN SPECIFIED.
3. Δ ANCHOR TYPE SHACKLE AND 2 1/4" LINK NOT REQUIRED WHEN TYPE B POLE TOP BAND, DWG NO. 41 6035 IS USED.
4. * WHEN AN ADDITIONAL INSULATOR DISC IS USED ON ANGLE STRUCTURES, AT RAILROAD OR MAIN HIGHWAY CROSSINGS, THIS DIMENSION SHALL BE INCREASED ONE FOOT.
5. Ø GREATER THAN 20-INCH LENGTH MAY BE REQUIRED FOR EXTRA LARGE POLES.

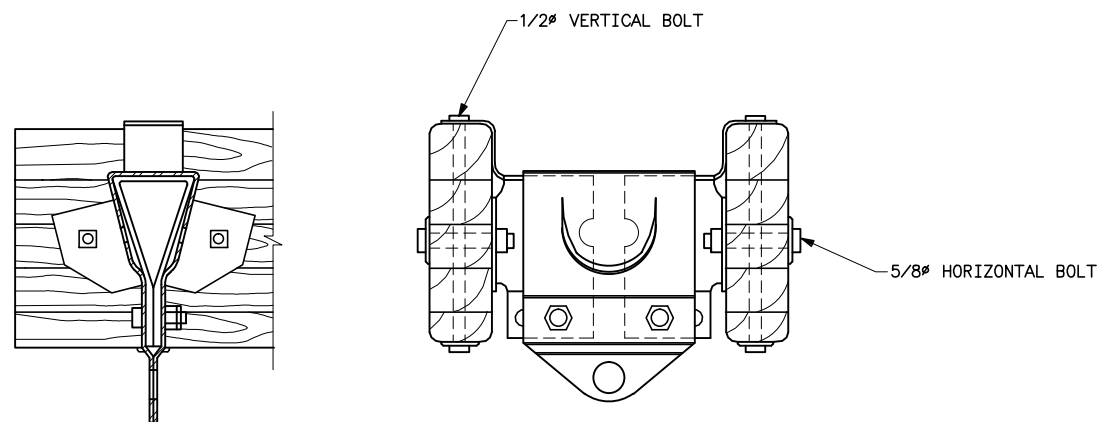
D	1-30-12 A7-KRR	CORRECTED A TYPO IN THE NOTES AND ADDED A MISSING PART NO.
C	6-14-11 A7-DDD	REDRAWN.
SUPERSEDES DWG NO. E40-D-4805 & 40-D-3802		
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGNS TRANSMISSION LINES TYPE HSB 69 AND 115KV SUSPENSION STRUCTURE		
DESIGNED BUREC	APPROVED BUREC	
AUGUST 6, 1979	41	6011



CROSSARM DRILLING DETAILS

NOTES

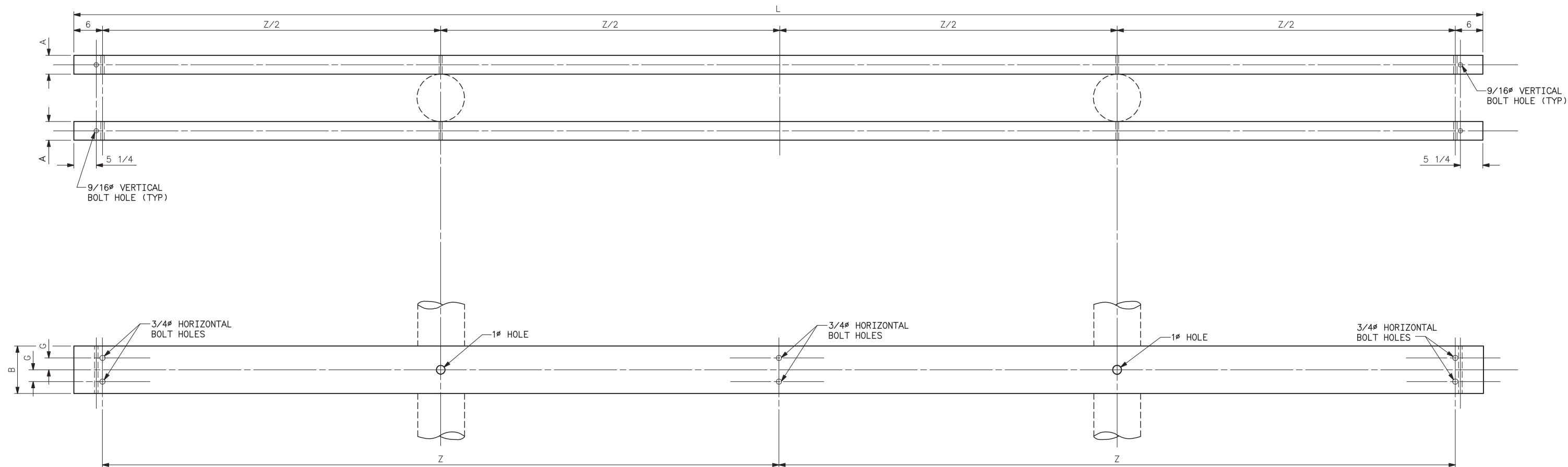
- CROSSARMS SHALL BE GLUED LAMINATED.
- ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS



DETAIL OF ADJUSTABLE SPACER FITTING

VOLTAGE	STRUCTURE TYPE	L	Z	Z/2	HUGHES BROS. SPACER FITTING (OR EQUAL) PART NO.
69KV	HS-1	21-0	10-0	5-0	3414.2
69KV	HSB-1	25-0	12-0	6-0	3414.2
115KV	HS-1 AND HSB-1	25-0	12-0	6-0	3414.2
138KV	HS-1 AND HSB-1	29-0	14-0	7-0	3415.13
161KV	HS-1 AND HSB-1	35-0	17-0	8-6	3415.13
230KV	HSB-1	45-0	22-0	11-0	3415.13

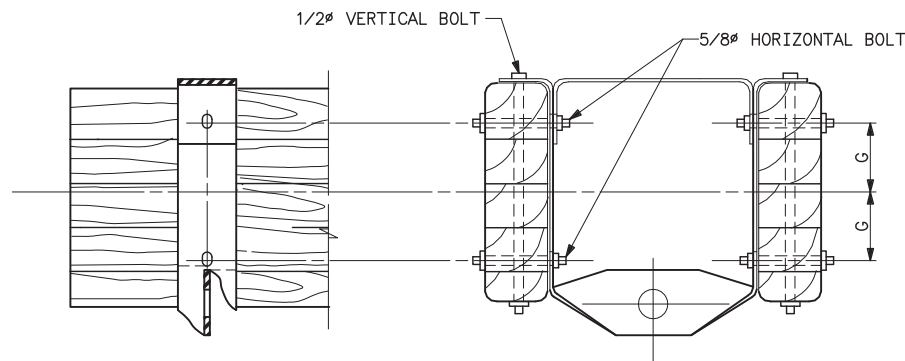
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - GOLDEN, COLORADO		
STANDARD DESIGNS TRANSMISSION LINES CROSSARM DRILLING DETAILS HS-1 AND HSB-1 STRUCTURES WITH ADJUSTABLE SPACER FITTINGS		
DESIGNED <u>KAREN ROWE</u>	APPROVED <u>DOUG HANSON</u> CIVIL ENGINEERING MANAGER	
AUGUST 30, 1996	41	6113



CROSSARM DRILLING DETAILS

NOTES

- CROSSARMS SHALL BE GLUED LAMINATED.
- THESE CROSSARMS SHALL BE USED ON EXISTING STRUCTURES WITH FIXED SPACER FITTINGS ONLY. ALL NEW STRUCTURES SHOULD USE STANDARD CROSSARM SIZES AND DETAILS SHOWN ON STANDARD DRAWING NUMBER 41 6113.

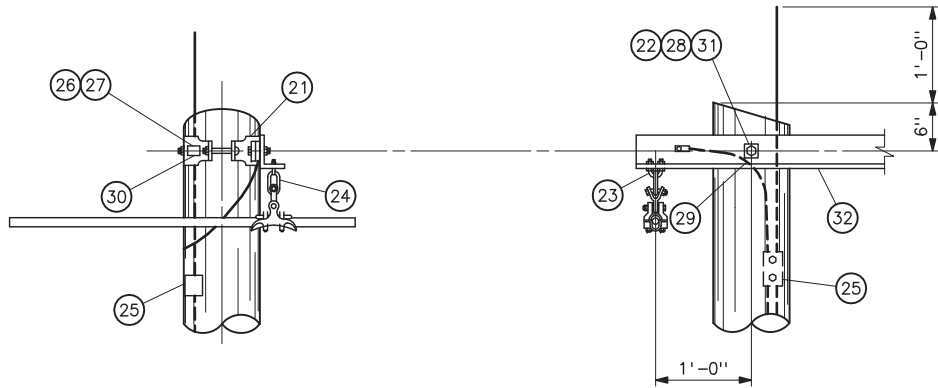
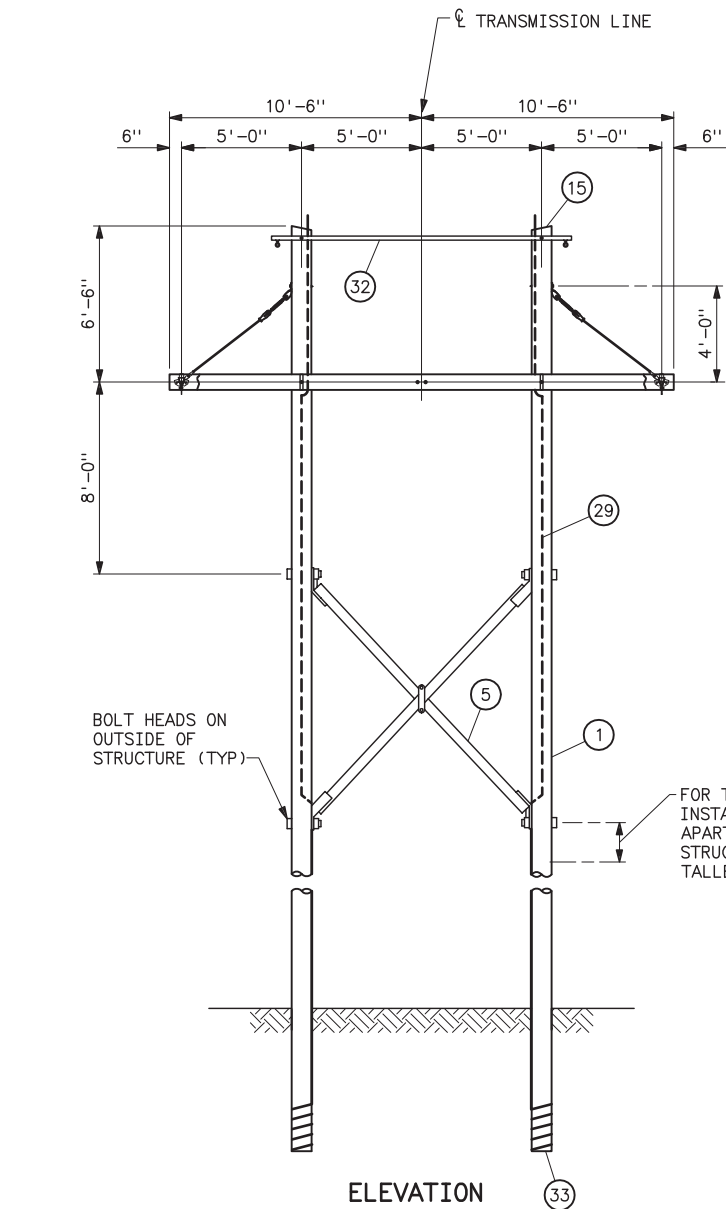


DETAIL OF FIXED SPACER FITTING
(STANDARD DESIGN PART NO. 40.49-40.68)

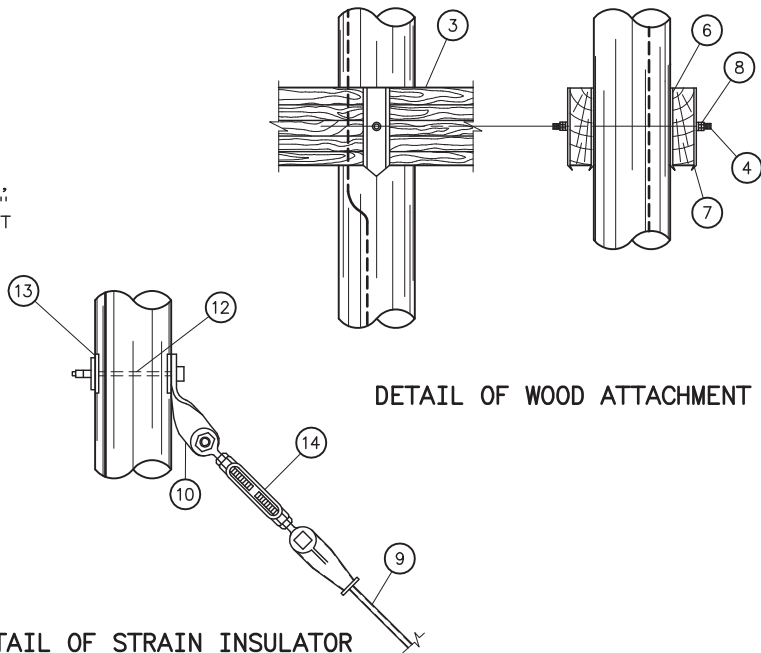
VOLTAGE	STRUCTURE TYPE	L	A	B	Z	Z/2	G	STANDARD DESIGN PART NO.	REFERENCE DRAWING NO.
69KV	HS	21-0	2 5/8	9 1/2	10-0	5-0	3 IN.	40.49-40.52	41 6038
69KV	HSB	25-0	2 5/8	9 1/2	12-0	6-0	3 IN.	40.49-40.52	41 6038
115KV	HS AND HSB	25-0	2 5/8	9 1/2	12-0	6-0	3 IN.	40.49-40.52	41 6038
138KV	HS AND HSB	29-0	3 5/8	11 1/2	14-0	7-0	3 3/4 IN.	40.53-40.56 40.60,40.61	41 6039
161KV	HS AND HSB	35-0	3 5/8	11 1/2	17-0	8-6	3 3/4 IN.	40.53-40.56 40.60,40.61	41 6039
230KV	HSB	45-0	3 1/8	9	22-0	11-0	2 1/2 IN.	40.66-40.68	41 6040

A	09-13-01 A7-KKR	ADDED NOTE 2. DELETED "-1" FROM TYPES.	
	SUPERSEDES DWG NO. 41 6023, 41 6025, 41 6027		
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION HEADQUARTERS - GOLDEN, COLORADO			
STANDARD DESIGNS TRANSMISSION LINES CROSSARM DRILLING DETAILS HS AND HSB STRUCTURES WITH FIXED SPACER FITTINGS			
DESIGNED	K.K. ROWE	RECOMMENDED	K.K. ROWE
DRAWN	K.K. ROWE	APPROVED	ROSS M. CLARK CHIEF, TRANSMISSION LINE BRANCH
CHECKED	K.K. ROWE		
DATE	APRIL 18, 1995	41	6114

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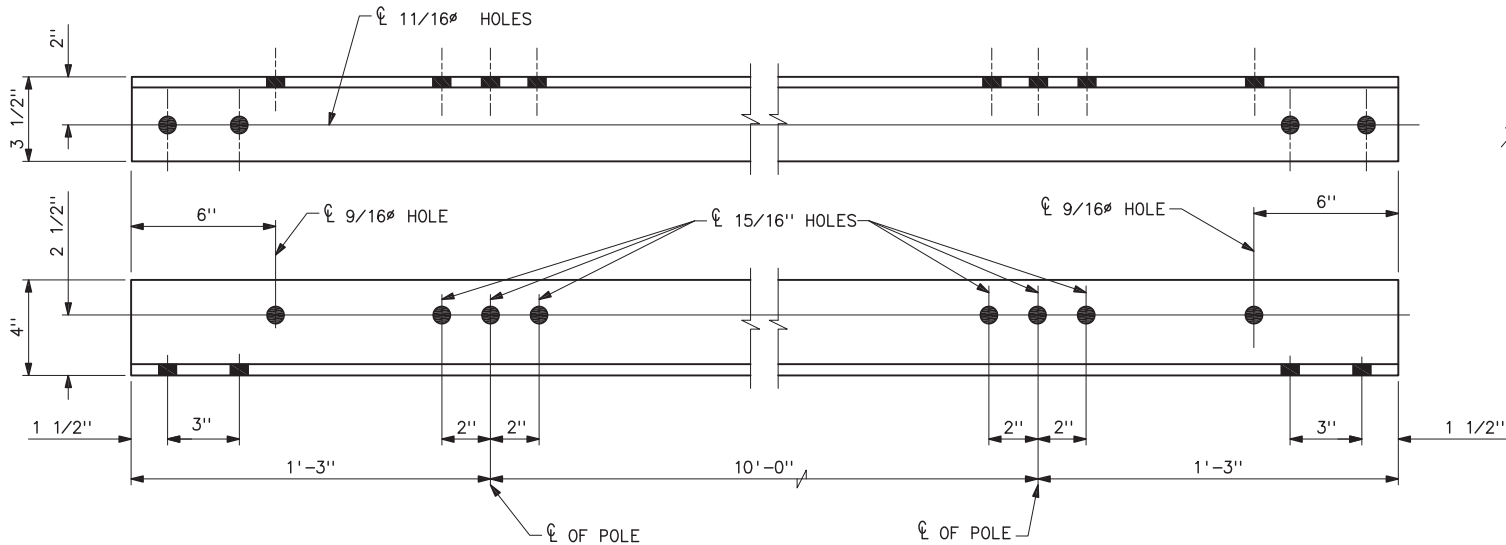


DETAIL OF OVERHEAD GROUND WIRE SUSPENSION ATTACHMENT

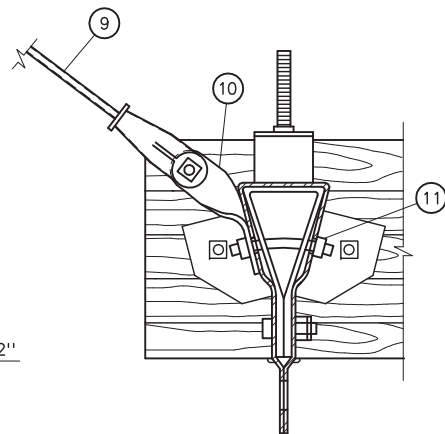


DETAIL OF WOOD ATTACHMENT

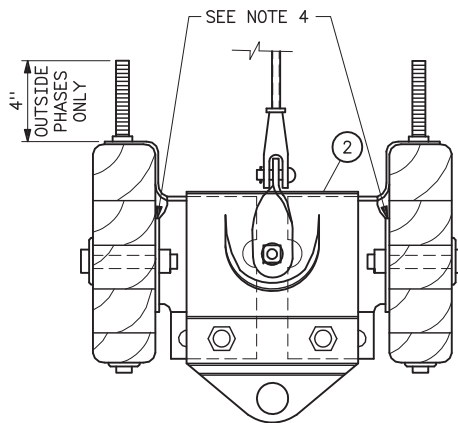
DETAIL OF STRAIN INSULATOR ATTACHMENT TO POLE



STEEL ANGLE (32)



DETAIL OF PHASE FITTING ASSEMBLY



STRUCTURE MATERIAL

PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	3	ADJUSTABLE SPACER FITTING, COMPLETE WITH BOLTS, NUTS AND WASHERS (HUGHES BROS. TYPE 3414 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8"x9"x21'-0"
4	2	DOUBLE ARMING BOLT, 7/8"x REQUIRED LENGTH WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 3 3/8"x4 3/8" (HUGHES BROS. NO. 1042 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8" BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 25,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (EQUAL TO HUGHES BROS. NO. AS2276)
11	2	7/8"x REQUIRED LENGTH BENT BOLT WITH LOCK WASHERS AND LOCKNUTS
12	2	7/8"x REQUIRED LENGTH BOLT WITH LOCK WASHER AND LOCKNUT
13	2	4x4x1/4 CURVED WASHER, 15/16" HOLE
14	2	TURNBUCKLE (HUGHES BROS. AS2545-B OR EQUAL)
15	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL

PART NO.	QUANTITY	DESCRIPTION
21	2	POLE TOP BAND (HUGHES BROS. NO. 2845-B OR EQUAL)
22	2	MACHINE BOLT, 7/8"x12"
23	2	"U" BOLT, 5/8"x3 WITH 4 HEX NUTS AND 2 LOCKNUTS
24	2	CHAIN LINK 1/2" (HUGHES BROS. NO. 1329-B1 OR EQUAL)
25	2	PARALLEL GROOVE CLAMP, 2 BOLT COPPER ALLOY OR COPPER CRIMPIT (BURNDY TYPE YC-C OR EQUAL)
26	2	1/2"x4" LAG SCREW WITH FETTER.DRIVE THREAD (ONE PER BAND)
27	2	1 3/8" FLAT ROUND WASHER WITH 9/16" HOLE (ONE PER BAND)
28	2	LOCKNUT FOR 7/8" BOLT
29	AS REQUIRED	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2" STAPLES
30	4	1/2" GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT).
31	2	SPRING LOCK WASHER FOR 7/8" BOLT
32	1	STEEL ANGLE, 4x3 1/2x12-6 LONG
33	2	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE

NOTES

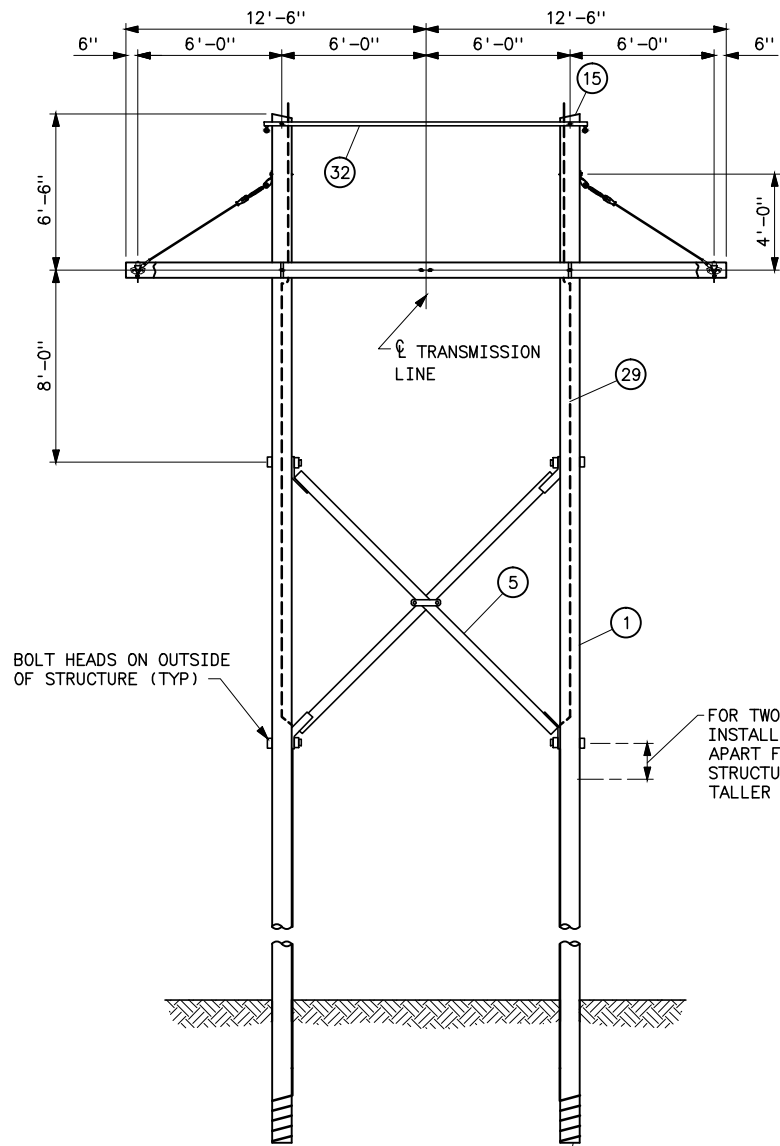
- STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
- ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
- GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
- HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16 INCH LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
- BNK INDICATES BOLT, NUT AND STAINLESS STEEL COTTER KEY REQUIRED.

REFERENCE DRAWINGS

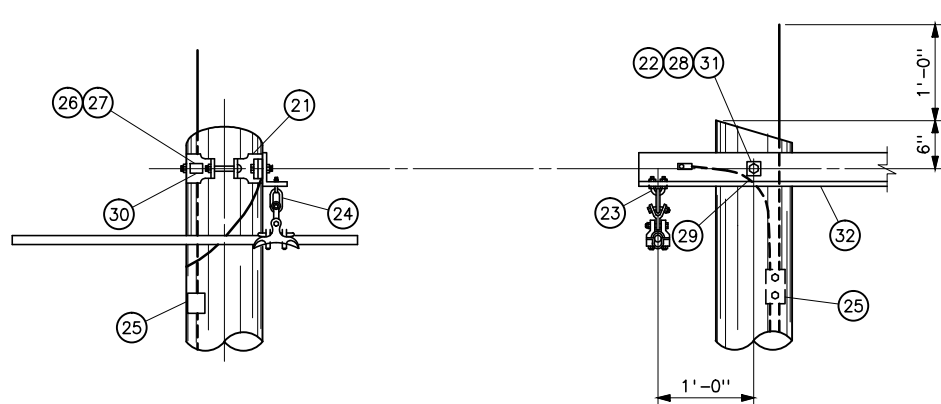
WOOD POLES STRUCTURES	
GROUNDING DETAILS	41 1012
OVERHEAD GROUND WIRE ASSEMBLY	
DETAILS FOR WOOD POLES STRUCTURES	41 1019
CROSSARM DRILLING DETAILS HS-1 AND	
HSB-1 STRUCTURES	41 6113
OPTICAL GROUNDWIRE ASSEMBLIES	41 1031
SUSPENSION INSULATOR ASSEMBLIES	
SINGLE STRING	41 1020

A	04-14-05 A7-KKR	MINOR CHANGES. ADDED STAR-LOCK.
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGN TRANSMISSION LINES TYPE HS-1 69KV SUSPENSION STRUCTURE		
DESIGNED KAREN ROWE APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER		
C.A.F.	MAY 16, 1997	41 6117

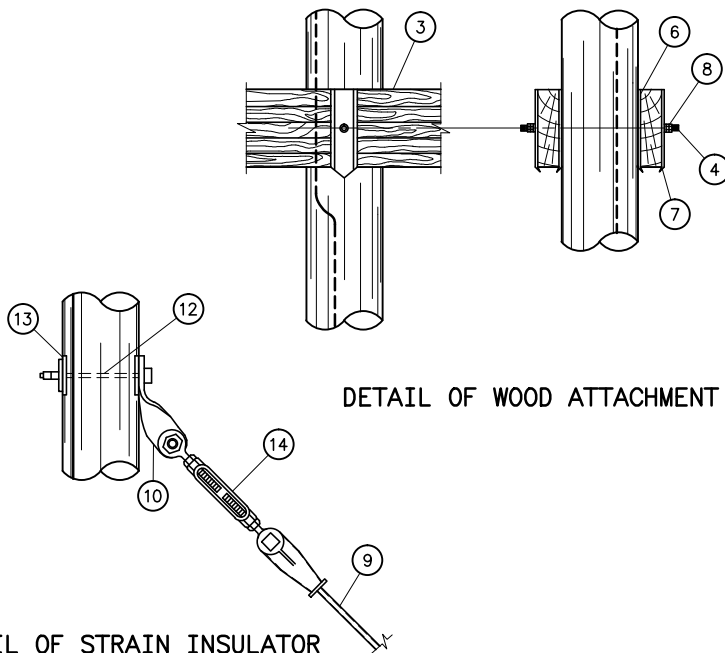
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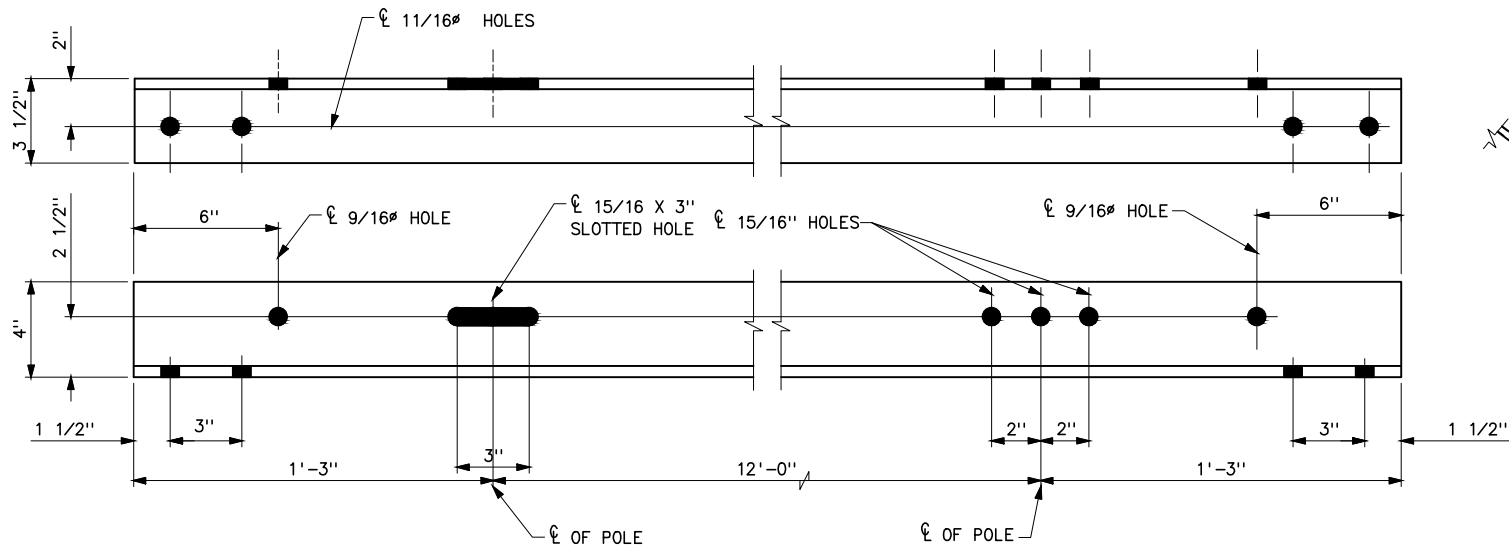


DETAIL OF OVERHEAD GROUND WIRE SUSPENSION ATTACHMENT

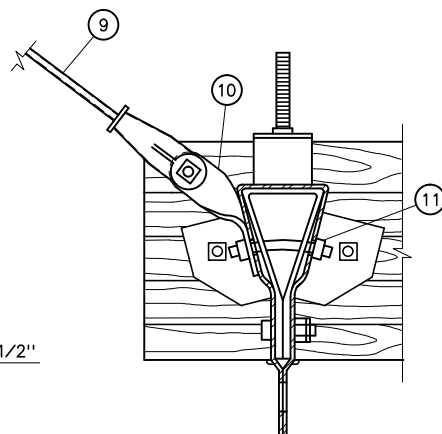


DETAIL OF WOOD ATTACHMENT

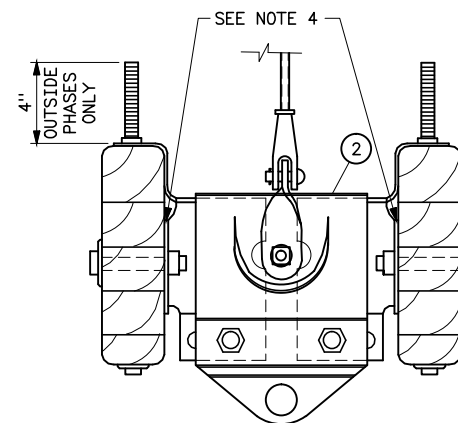
DETAIL OF STRAIN INSULATOR ATTACHMENT TO POLE



STEEL ANGLE (32)



DETAIL OF PHASE FITTING ASSEMBLY



STRUCTURE MATERIAL

PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	3	ADJUSTABLE SPACER FITTINGS COMPLETE WITH BOLTS, NUTS, & WASHERS (HUGHES BROS. TYPE 3414.2 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8"x9"x25'-0"
4	2	DOUBLE ARMING BOLT, 7/8"x REQD LENGTH WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 3 3/8"x4 3/8" (HUGHES BROS. NO. 1042 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8" BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 25,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (EQUAL TO HUGHES BROS NO. AS2276)
11	2	7/8"x REQD LENGTH BENT BOLT WITH LOCK WASHERS AND LOCKNUTS
12	2	7/8"x REQD LENGTH BOLT WITH LOCK WASHER AND LOCKNUT
13	2	4"x4"x1/4" CURVED WASHER, 15/16" HOLE
14	2	TURNBUCKLE (HUGHES BROS NO. AS2545-B OR EQUAL)
15	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL

PART NO.	QUANTITY	DESCRIPTION
21	2	POLE TOP BAND (HUGHES BROS. NO. 2845-B OR EQUAL)
22	2	MACHINE BOLT, 7/8"x1'-0"
23	2	"U" BOLT, 5/8"x3" WITH 4 HEX NUTS AND 2 LOCKNUTS
24	2	CHAIN LINK 1/2" (HUGHES BROS NO. 1329-B1 OR EQUAL)
25	2	PARALLEL GROOVE CLAMP, 2-BOLT COPPER ALLOY OR COPPER CRIMPIT (BURNDY TYPE YC-C OR EQUAL)
26	2	1/2"x4" LAG SCREW WITH FETTER DRIVE THREAD (ONE PER BAND)
27	2	1 3/8" FLAT ROUND WASHER WITH 9/16" HOLE (ONE PER BAND)
28	2	LOCKNUT FOR 7/8" BOLT
29	AS REQUIRED	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2" STAPLES
30	4	1/2" GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT).
31	2	SPRING LOCK WASHER FOR 7/8" BOLT
32	1	STEEL ANGLE, 4"x3 1/2"x5/16"x 14'-6" LONG
33	2	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE

NOTES

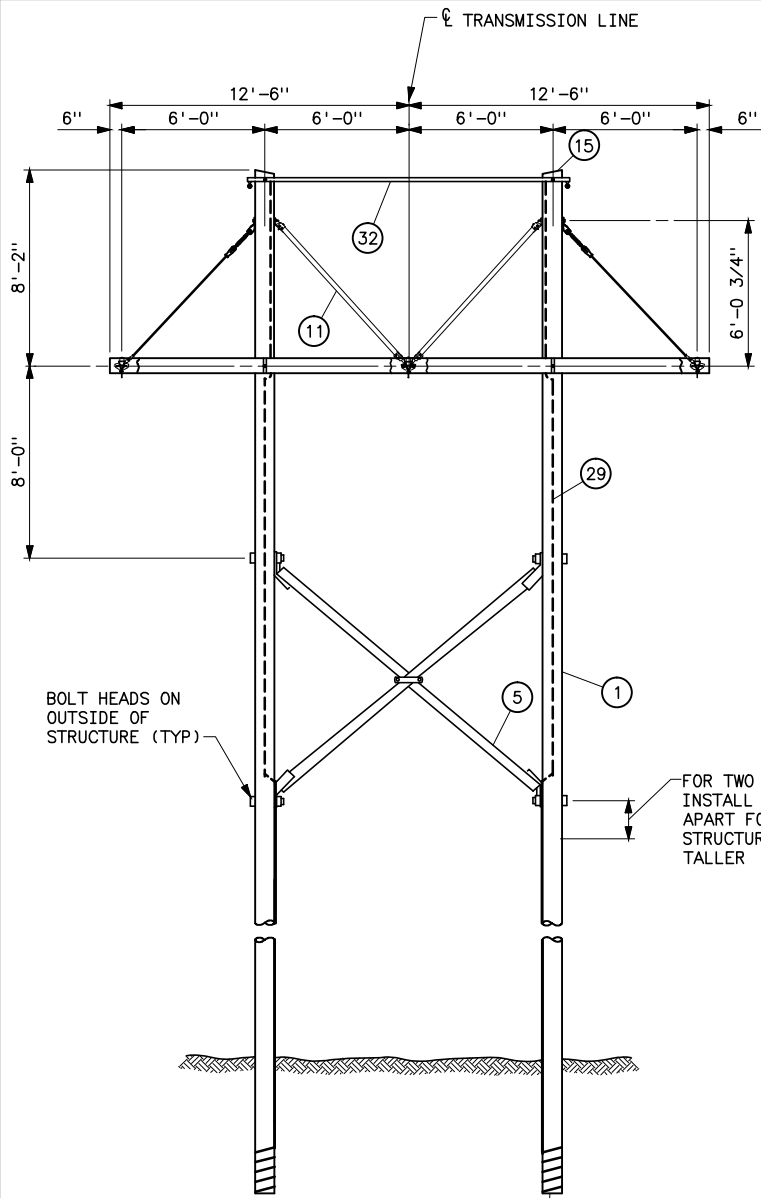
- STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
- ALL HOLES IN WOOD SHALL BE BORED 1/8" LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
- GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
- HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16" LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
- BNK INDICATES BOLT, NUT, AND STAINLESS STEEL COTTER KEY REQUIRED.

REFERENCE DRAWINGS

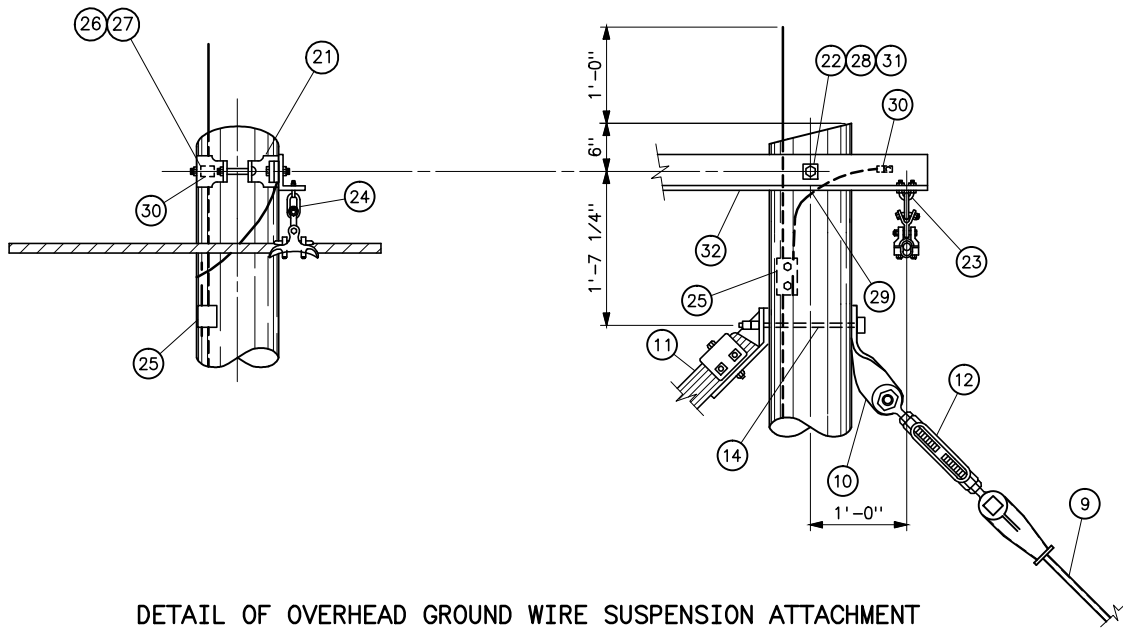
WOOD POLES STRUCTURES	
GROUNDING DETAILS	41 1012
OVERHEAD GROUND WIRE ASSEMBLY	
DETAILS FOR WOOD POLES STRUCTURES	41 1019
CROSSARM DRILLING DETAILS HS-1 AND	
HSB-1 STRUCTURES	41 6113
OPTICAL GROUNDWIRE ASSEMBLIES	41 1031
SUSPENSION INSULATOR ASSEMBLIES	
SINGLE STRING	41 1020

B	3-6-06 A7-KKR	MINOR REVISIONS
A	01-06-05 A7-KKR	MINOR CHANGES. ADDED STAR-LOCK.
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGNS TRANSMISSION LINES TYPE HS-1 115KV SUSPENSION STRUCTURE		
DESIGNED KAREN ROWE APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER		
CAE	AUGUST 30, 1996	41 6122

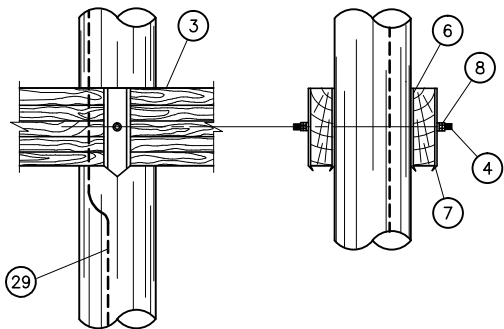
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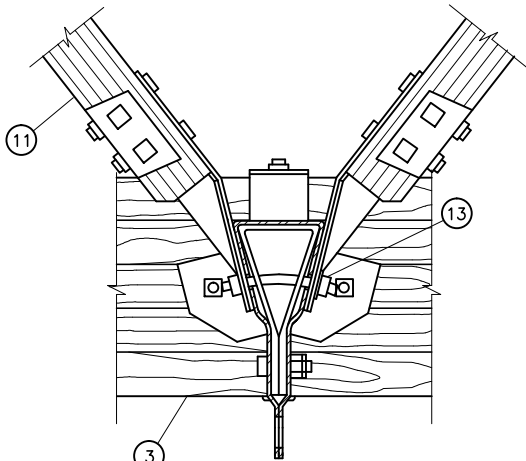
ELEVATION



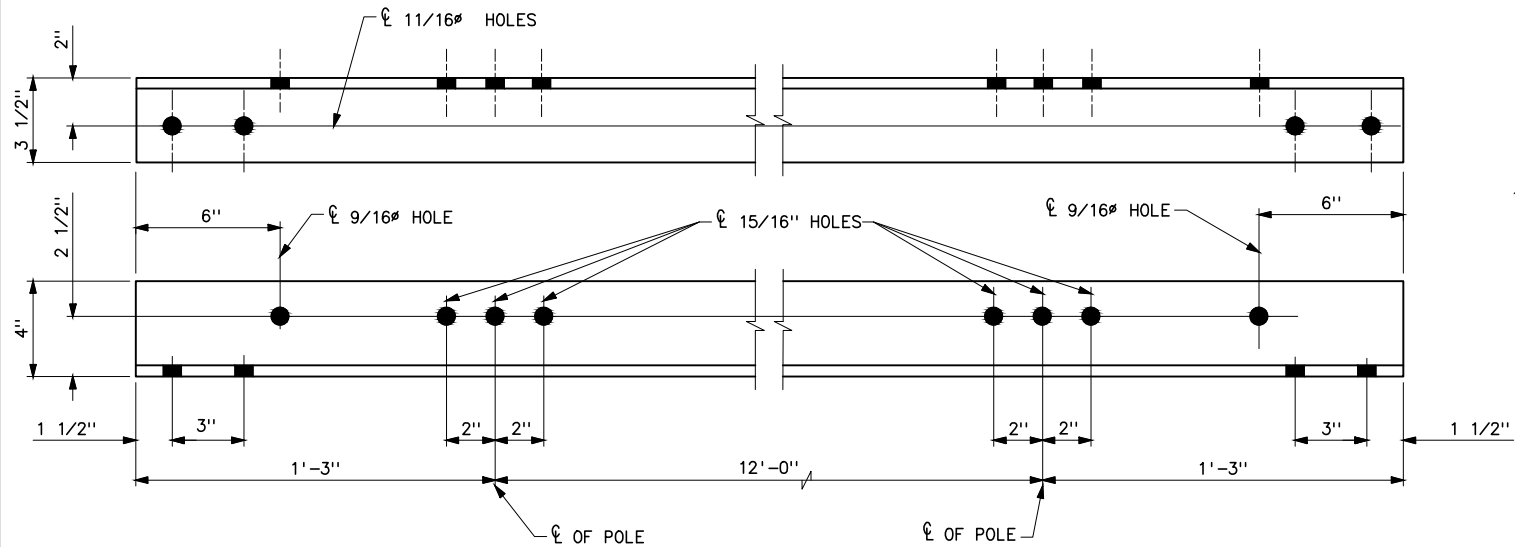
DETAIL OF OVERHEAD GROUND WIRE SUSPENSION ATTACHMENT AND BRACE ATTACHMENTS TO POLE



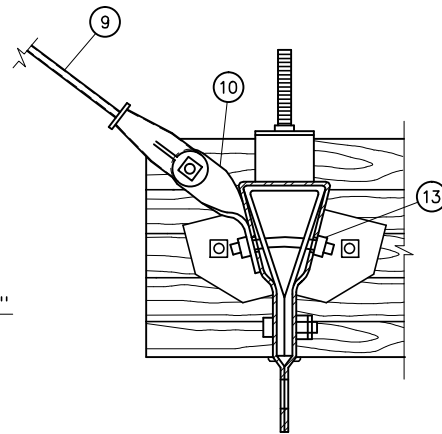
DETAIL OF WOOD ATTACHMENT



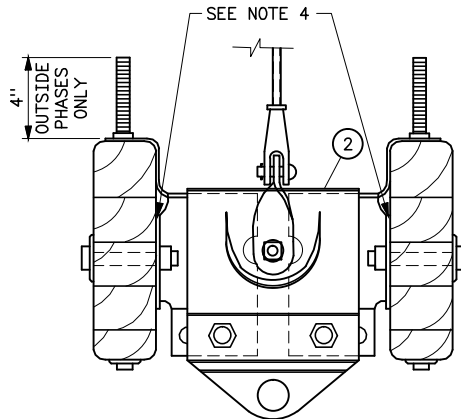
DETAIL OF CENTER PHASE FITTING ATTACHMENTS



STEEL ANGLE (32)



DETAIL OF END PHASE FITTING ASSEMBLY



STRUCTURE MATERIAL

PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	3	ADJUSTABLE SPACER FITTINGS COMPLETE W/BOLTS, NUTS, & WASHERS (HUGHES BROS. TYPE 3414 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8"x9"x25'-0"
4	2	DOUBLE ARMING BOLT, 7/8"Ø x REQD LENGTH W/4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 3 3/8"x4 3/8" (HUGHES BROS. NO. 1042 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8"Ø BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 25,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (HUGHES BROS NO. AS2276 OR EQUAL)
11	2	WOOD VEE BRACE, 3 3/8"x4"
12	2	TURNBUCKLE (HUGHES BROS. NO. AS2545-B OR EQUAL)
13	3	7/8"ØxREQD LENGTH BENT STUD WITH LOCK WASHERS AND LOCKNUTS, THREADED BOTH SIDES
14	2	7/8"ØxREQD LENGTH BOLT WITH LOCK WASHER AND LOCKNUT
15	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL

PART NO.	QUANTITY	DESCRIPTION
21	2	POLE TOP BAND (HUGHES BROS. NO. 2845-B OR EQUAL)
22	2	MACHINE BOLT, 7/8"Øx1'-0"
23	2	"U" BOLT, 5/8"Øx3" WITH 4 HEX NUTS AND 2 LOCKNUTS
24	2	CHAIN LINK 1/2"Ø (HUGHES BROS. NO. 1329-B1 OR EQUAL)
25	2	PARALLEL GROOVE CLAMP, 2-BOLT COPPER ALLOY OR COPPER CRIMPIT (BURNDY TYPE YC-C OR EQUAL)
26	2	1/2"Øx4" LAG SCREW WITH FETTER DRIVE THREAD (ONE PER BAND)
27	2	1 3/8"Ø FLAT ROUND WASHER WITH 9/16"Ø HOLE (ONE PER BAND)
28	2	LOCKNUT FOR 7/8"Ø BOLT
29	AS REQUIRED	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2" STAPLES
30	4	1/2"Ø GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT AND EQUAL TO HUGHES BROS. NO 2718.55)
31	2	SPRING LOCK WASHER FOR 7/8"Ø BOLT
32	1	STEEL ANGLE, 4x3 1/2x5/16x14'-6" LONG
33	2	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE

NOTES

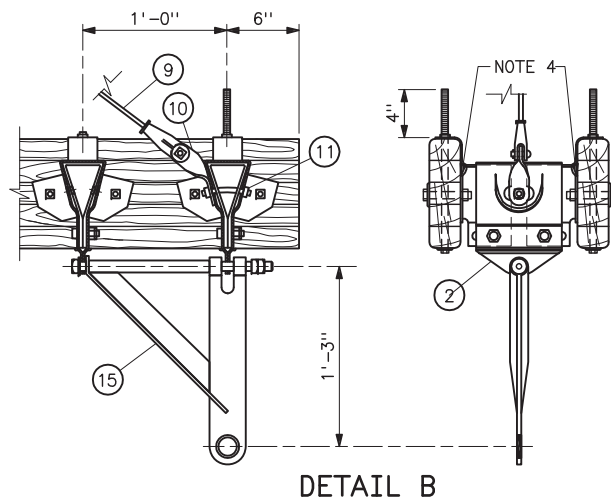
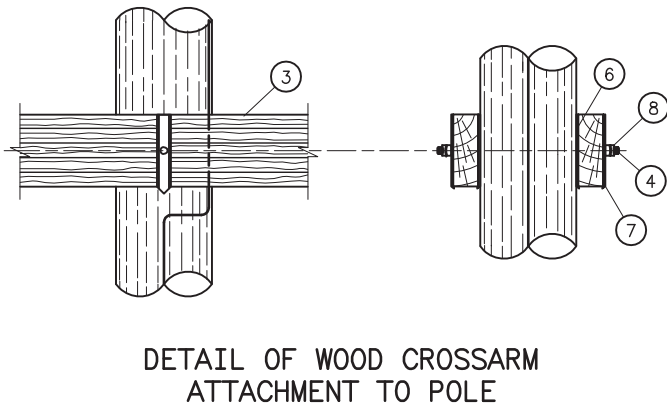
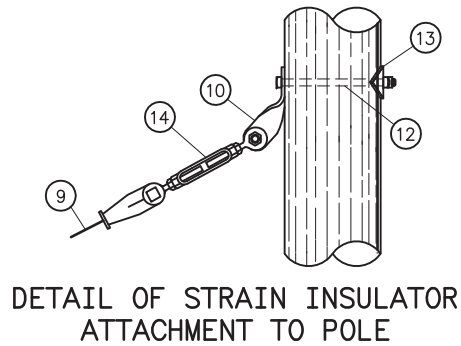
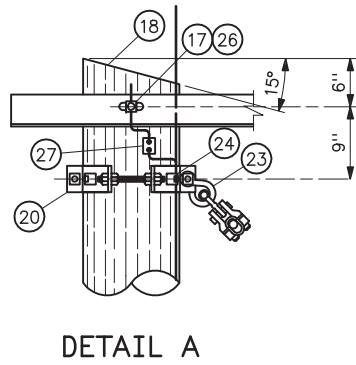
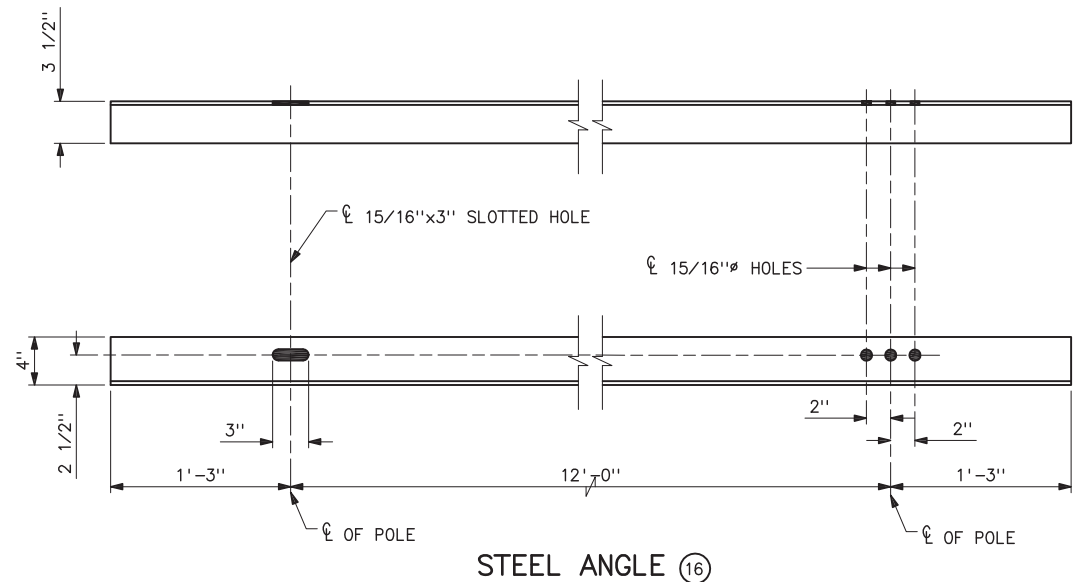
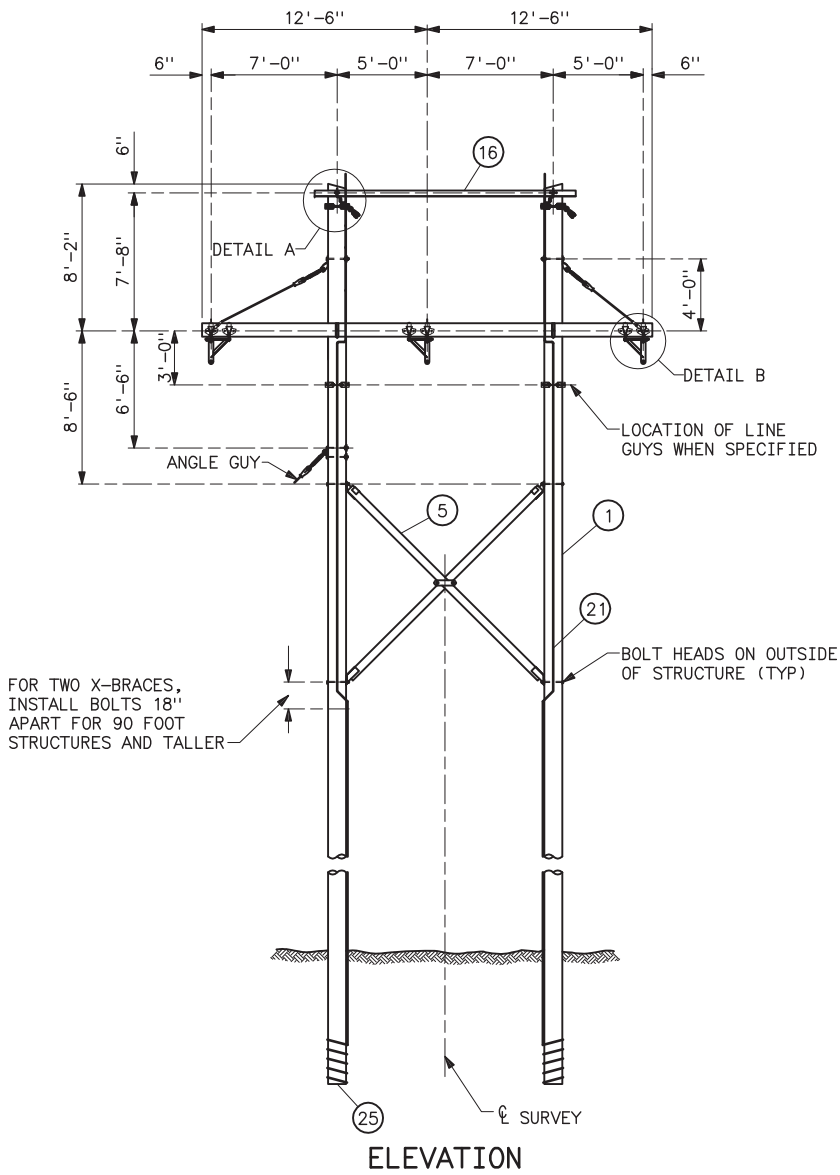
- STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
- ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS
- GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
- HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16 INCH LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
- BNK INDICATES BOLT, NUT, AND STAINLESS STEEL COTTER KEY REQUIRED.

REFERENCE DRAWINGS

WOOD POLES STRUCTURES
GROUNDING DETAILS-----41 1012
OVERHEAD GROUND WIRE ASSEMBLY
DETAILS FOR WOOD POLES STRUCTURES-----41 1019
CROSSARM DRILLING DETAILS HS-1 AND
HSB-1 STRUCTURES-----41 6113
OPTICAL GROUNDWIRE ASSEMBLIES-----41 1031
SUSPENSION INSULATOR ASSEMBLIES
SINGLE STRING-----41 1020

B	1-30-12 A7-KRR	CORRECTED PART NUMBER IN 'DETAIL OF END PHASE FITTING ASSEMBLY'.
A	04-14-05 A7-KRR	MINOR CHANGES. ADDED STAR-LOCK.
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGN TRANSMISSION LINES TYPE HSB-1 69 AND 115KV SUSPENSION STRUCTURE		
DESIGNED KAREN ROWE APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER		
CAE	AUGUST 30, 1996	41 6123

Jun 13, 2006- 8:19am Plotted By: Seela IMAGES:
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STRUCTURE MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	6	ADJUSTABLE SPACER FITTING, COMPLETE WITH BOLTS, NUTS AND WASHERS (HUGHES BROS. TYPE 3414 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8x9x25-0
4	2	DOUBLE ARMING BOLT, 7/8" x REQUIRED LENGTH WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 3 3/8x4 3/8 (HUGHES BROS. NO. 1042 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8" BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 25,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (EQUAL TO HUGHES BROS. NO. AS2276)
11	2	7/8" x REQUIRED LENGTH BENT BOLT WITH LOCK WASHERS AND LOCKNUTS
12	2	7/8" x REQUIRED LENGTH BOLT WITH LOCK WASHER AND LOCKNUT
13	2	4x4x1/4 CURVED WASHER, 15/16" HOLE
14	2	TURNBUCKLE (HUGHES BROS. AS2545-B OR EQUAL)
15	3	SWINGING ANGLE BRACKET, COMPLETE WITH FITTINGS (HUGHES BROS. NO. 2821.6 OR EQUAL)
16	1	STEEL ANGLE, 4x3 1/2x5/16x14-6 LONG
17	2	MACHINE BOLT, 7/8" x REQUIRED LENGTH WITH LOCKNUT, SPRING LOCK WASHER AND 4x4x1/4 CURVED WASHER ON BACK SIDE
18	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY.

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
20	2	FOUR-WAY STRAIN POLE BAND (EQUAL TO HUGHES BROS. 3103 SERIES) (SEE NOTE 6)
	4	1/2"x4 LAG SCREWS (2 PER BAND)
	4	1 3/8" FLAT ROUND WASHER WITH 9/16" HOLE (2 PER BAND)
21	AS REQD	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2" STAPLES
23	2	TWISTED SHACKLE (EQUAL TO HUGHES BROS. NO. 2866)
24	2	1/2" GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT). (EQUAL TO HUGHES BROS. 2718.55)
25	2	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE
26	2	BONDING CLIP (EQUAL TO HUGHES BROS. NO. 2727)
27	2	PARALLEL GROOVE CLAMP, 2-BOLT COPPER ALLOY

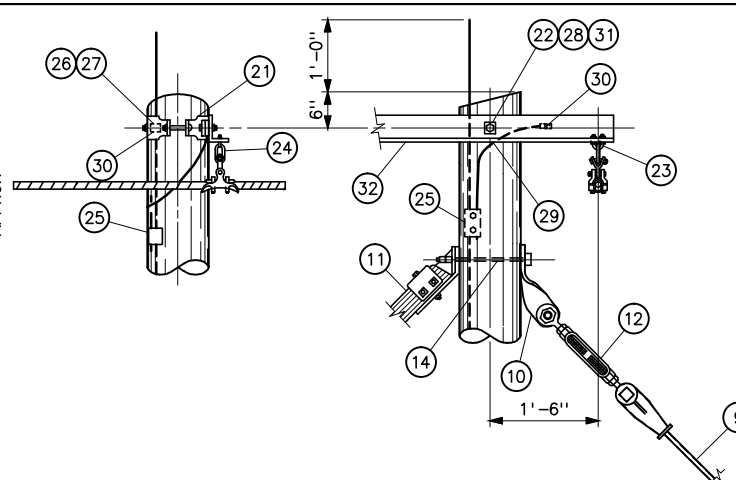
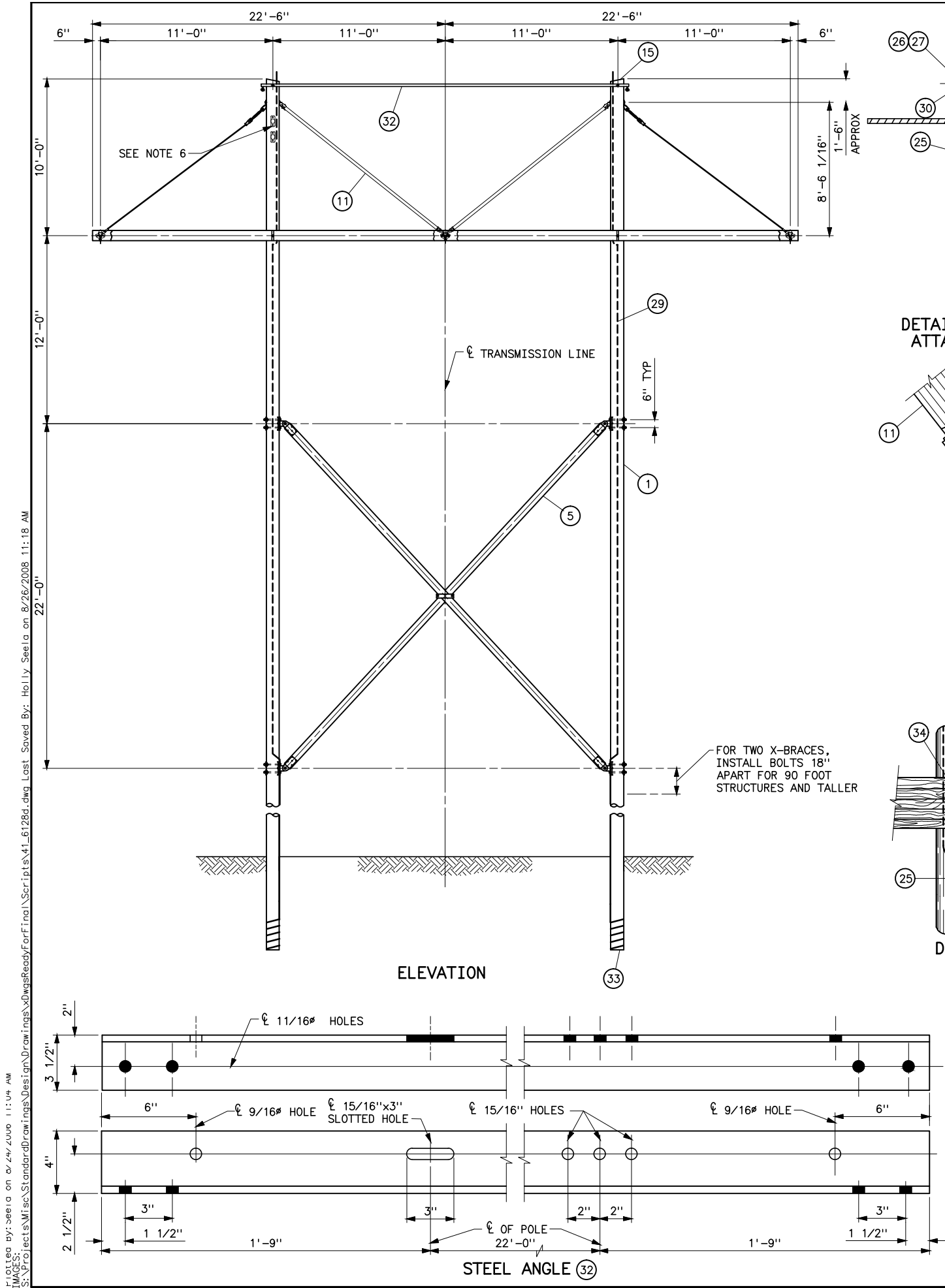
NOTES

1. STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
2. ALL HOLES IN WOOD SHALL BE BORED 1/8" LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
3. GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
4. HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16" LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
5. BNK INDICATES BOLT, NUT AND STAINLESS STEEL COTTER KEY REQUIRED.
6. DEAD END TEES MAY BE USED IN LIEU OF POLE BANDS. CONNECTING LINKS, SHEAVE WHEELS AND GROUNDING CLIP ASSEMBLIES ARE NOT REQUIRED WITH DEAD END TEES. SEE STANDARD DRAWING 41 6138.

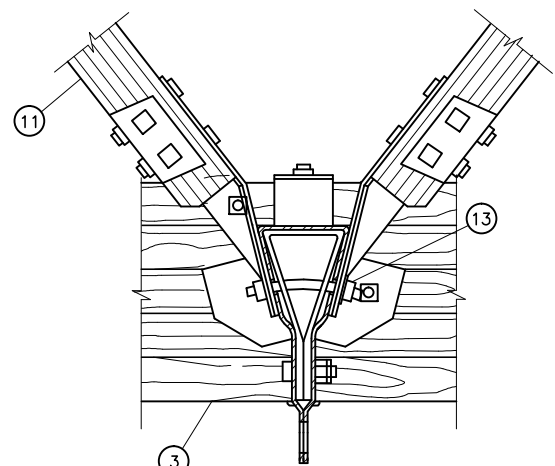
REFERENCE DRAWINGS

WOOD POLE STRUCTURES
GROUNDING DETAILS _____ 41 1012
WOOD POLE STRUCTURES
OVERHEAD GROUND WIRE ASSEMBLY _____ 41 1019
SUSPENSION INSULATOR ASSEMBLIES
SINGLE STRING _____ 41 1020
OPTICAL GROUND WIRE ASSEMBLIES _____ 41 1031
GUY ASSEMBLIES AND DETAILS _____ 41 6047
STANDARD GUY ARRANGEMENT, 2 POLE
ANGLE STRUCTURES _____ 41 6052
TYPICAL DETAILS FOR DEAD-END TEES _____ 41 6138
CROSSARM DRILLING DETAILS -
69 AND 115KV HA-1 STRUCTURE _____ 41 6139

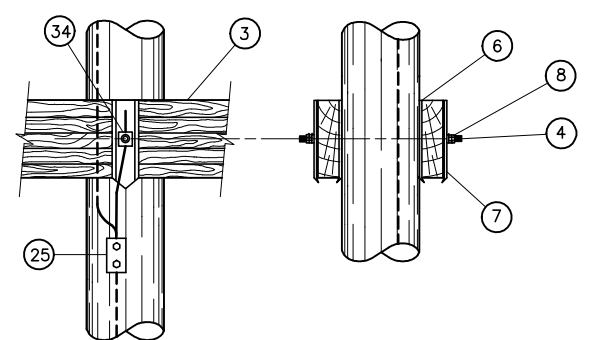
B	09-27-05 A7-KKR	MINOR REVISION, MOVED STEEL ANGLE TO TOP OF STRUCTURE AND MODIFIED STEEL ANGLE.
A	02-07-05 A7-KKR	REVISED DRAWING LIST. ADDED STAR-LOCK AND NOTE 6.
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGN TRANSMISSION LINES TYPE HA-1 69 AND 115KV SMALL ANGLE SUSPENSION STRUCTURE		
DESIGNED	KAREN ROWE	APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER
DATE	MAY 01, 1997	41 6124



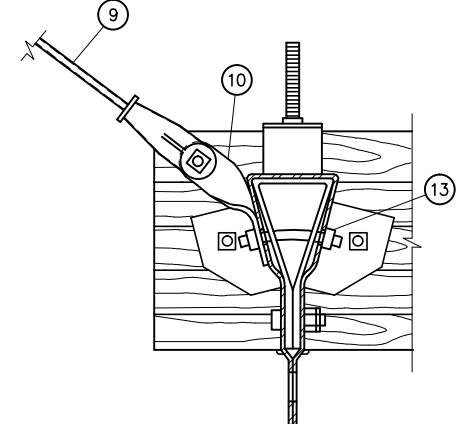
DETAIL OF OVERHEAD GROUND WIRE SUSPENSION ATTACHMENT AND BRACE ATTACHMENT TO POLE



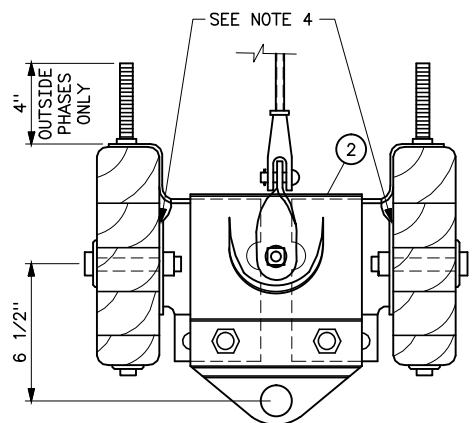
DETAIL OF CENTER PHASE FITTING ATTACHMENTS



DETAIL OF WOOD CROSSARM ATTACHMENT TO POLE



DETAIL OF PHASE FITTING ASSEMBLY



STRUCTURE MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	3	ADJUSTABLE SPACER FITTINGS COMPLETE WITH BOLTS, NUTS, AND WASHERS (HUGHES BROS. TYPE 3415 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8x9x45-0
4	2	DOUBLE ARMING BOLT, 7/8"x REQUIRED LENGTH WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 5 1/8x6 (HUGHES BROS. NO. 2121-22-0 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8" BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 30,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (HUGHES BROS. NO. AS2276 OF EQUAL)
11	2	WOOD VEE BRACE, 3 1/2x7 1/2, COMPLETE WITH FITTINGS, 8'-6" DROP, SPACING VARIES WITH POLE DIAMETER
12	2	TURNBUCKLE (HUGHES BROS. NO. AS2545-B OF EQUAL)
13	3	7/8"x REQUIRED LENGTH BENT STUD WITH LOCK WASHER AND LOCKNUTS, THREADED BOTH SIDES
14	2	7/8"x REQUIRED LENGTH WITH LOCK WASHER AND LOCKNUT
15	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR LOCK MANUFACTURED BY THE BAYNE COMPANY

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
21	2	POLE TOP BAND (HUGHES BROS. NO. 2845-B OR EQUAL)
22	2	MACHINE BOLT, 7/8"x1-0
23	2	"U" BOLT, 5/8"x3 WITH 4 HEX NUTS AND 2 LOCKNUTS
24	2	CHAIN LINK 1/2" (HUGHES BROS. NO. 1329-B1 OR EQUAL)
25	4	PARALLEL GROOVE CLAMP, 2 BOLT COPPER ALLOY OR COPPER CRIMPIT (BURNDY TYPE YC-C OR EQUAL)
26	2	1/2"x4 LAG SCREW WITH FETTER DRIVE THREAD (ONE PER BAND)
27	2	1 3/8" FLAT ROUND WASHER WITH 9/16" HOLE (ONE PER BAND)
28	2	LOCKNUT FOR 7/8" BOLT
29	AS REQUIRED	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2 INCH STAPLES
30	3	1/2" GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT)
31	2	SPRING LOCK WASHER FOR 7/8" BOLT
32	1	STEEL ANGLE, 4x3 1/2x3/8x25-6 LONG
33	2	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE
34	2	BONDING CLIP (HUGHES BROS. NO. 2727 OR EQUAL)

NOTES

- STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
- ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
- GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
- HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16 LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
- BNK INDICATES BOLT, NUT, AND STAINLESS STEEL COTTER KEY REQUIRED.
- FOR DEAD ENDING OPGW, SPACE DEAD END TEE TOP BOLT 6" BELOW BRACE ATTACHMENT.

REFERENCE DRAWINGS

WOOD POLE STRUCTURE GROUNDING DETAILS	41 1012
OVERHEAD GROUND WIRE ASSEMBLY DETAILS	
FOR WOOD POLE STRUCTURES	41 1019
SUSPENSION INSULATOR ASSEMBLIES	
SINGLE STRING	41 1020
OPTICAL GROUNDWIRE ASSEMBLIES	41 1031
CROSSARM DRILLING DETAILS HS-1 AND	
HSB-1 STRUCTURES	41 6113
TYPICAL DETAILS FOR OPTICAL GROUND	
WIRE SPLICE STRUCTURES	41 6153

D	8-26-08 A7-KKR	CHANGED QUANTITY OF ITEM 33 FROM 1 TO 2.
C	08-24-06 A7-KKR	CHANGED QUANTITY OF PART NO. 32 FROM 2 TO 1.

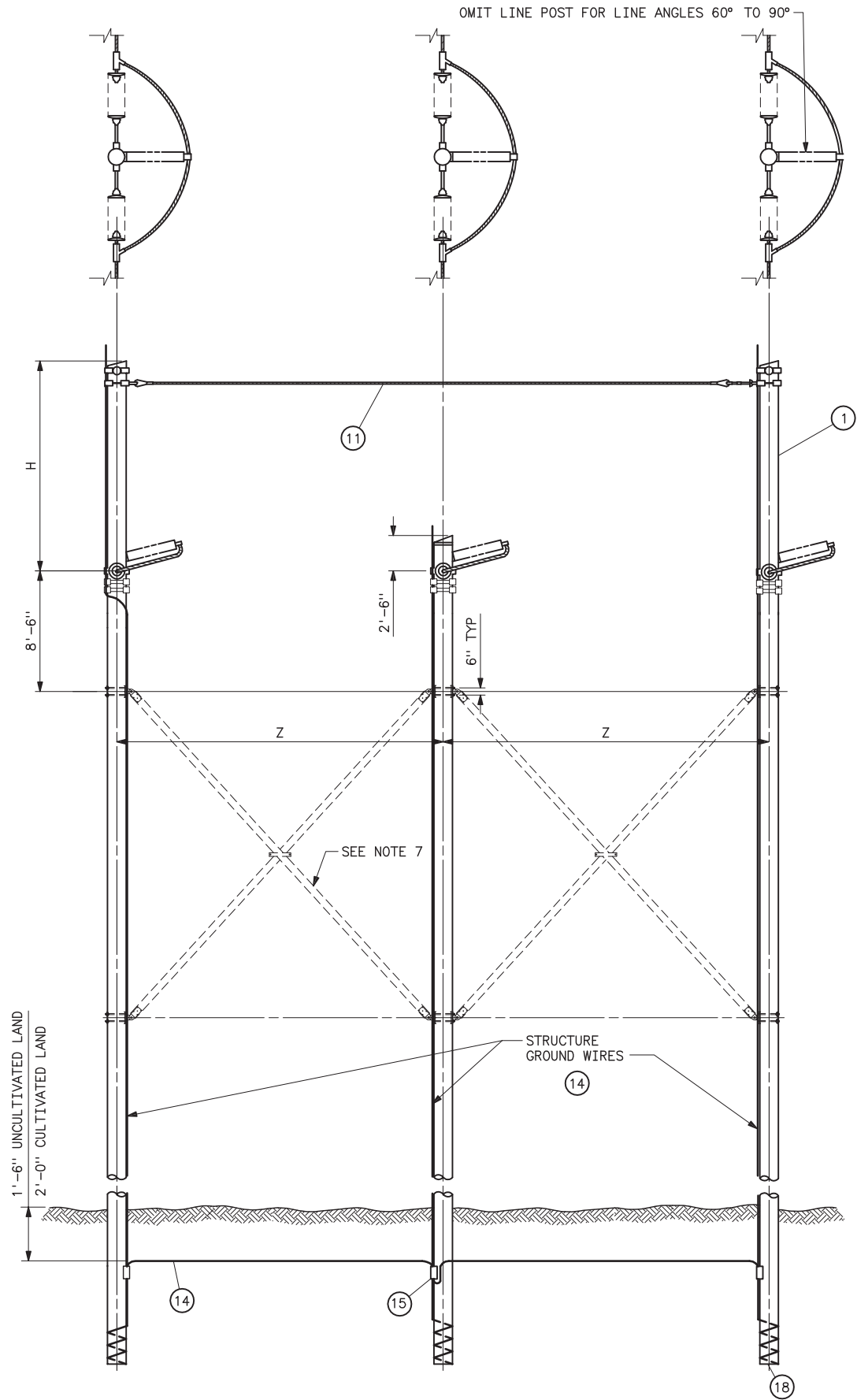
UNITED STATES DEPARTMENT OF ENERGY
WESTERN AREA POWER ADMINISTRATION
CORPORATE SERVICES OFFICE - GOLDEN, COLORADO

STANDARD DESIGNS
TRANSMISSION LINES
TYPE HSB-1 230KV
SUSPENSION STRUCTURE

DESIGNED KAREN ROWE APPROVED DOUG HANSON
CIVIL ENGINEERING MANAGER

February 7, 1997	41	6128
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VOLTAGE	Z				H	
	ANGLE TURNED IN DEGREES					
	0°-30°	30°-45°	45°-60°	60°-90°	0°-60°	60°-90°
230KV	23'-0"	28'-0"	35'-0"	48'-0"	20'-6"	18'-6"

STRUCTURE MATERIAL					
PART NO.	ANGLE	ANGLE	ANGLE	ANGLE	DESCRIPTION
	0°-5°	5°-30°	30°-60°	60°-90°	
1	3	3	3	3	POLE, SEE STRUCTURE LIST FOR LENGTH
2	—	6	6	6	FOUR-WAY STRAIN POLE BAND; HUGHES 3112 SERIES, JOSLYN J25969 SERIES; OR, APPROVED EQUAL
3	7	11	7	6	FOUR-WAY STRAIN POLE BAND; HUGHES 3103 SERIES, JOSLYN J25967 SERIES; OR, APPROVED EQUAL
4	3	—	—	—	POLE BAND ASSEMBLY, HUGHES B1784-VV OR APPROVED EQUAL
5	6	6	6	—	FIBERGLASS GUY INSULATOR, 10 FT LENGTH
6	—	6	6	6	CONNECTING LINK; PAIR W/BNK HUGHES 3157, JOSLYN J26035; OR, APPROVED EQUAL
7	6	6	6	6	CONNECTING LINK, PAIR W/BNK; HUGHES 3154; JOSLYN J6650; OR, APPROVED EQUAL CONNECTING LINK W/BNK
8	2	2	2	2	DEAD-END AUTOMATIC VISE TYPE FOR GUY STRAND
9	2	2	2	2	SHEAVE WHEEL, 2 1/2 DIA MINIMUM
10	1	1	1	1	TURNBUCKLE, CLEVIS-EYE, 7/8x12
11	AS REQD	AS REQD	AS REQD	AS REQD	STEEL STRAND 7/16, 7-WIRE, HIGH STRENGTH
12	7	17	13	12	1/2# GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT). HUGHES 2718.55 OR EQUAL
13	14	22	14	12	LAG SCREWS, 1/2x4 WITH FLAT WASHERS (2 EACH PER BAND)
14	AS REQD	AS REQD	AS REQD	AS REQD	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2-INCH STAPLES
15	3	3	3	3	PARALLEL GROOVE CLAMP, 2-BOLT COPPER ALLOY
16	4	4	4	4	OVERHEAD GROUND WIRE TENSION ASSEMBLY
17	6	—	—	—	CLEVIS-CLEVIS 90° W/BNK, 40,000 LBS
18	3	3	3	3	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE
19	3	3	3	3	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY

POST INSULATOR ASSEMBLY MATERIAL
(OMIT FOR 60° TO 90° ANGLES)

PART NO.	QUANTITY	DESCRIPTION
	0°-60°	
20	6	7/8xREQD LENGTH MACHINE BOLT WITH LOCKNUT AND LOCK WASHER
21	6	4x4x1/4 CURVED SQUARE WASHER WITH 15/16" HOLE
22	3	BONDING CLIP (EQUAL TO BROOKS MANU BC SERIES OR HUGHES BROS NO. 2727)

NOTES

- FITTINGS SHALL INCLUDE BOLTS, NUTS, AND WASHERS.
- ALL HOLES SHALL BE BORED 1/8 INCH LARGER IN WOOD POLE THAN THEIR RESPECTIVE BOLT DIAMETERS.
- MATERIAL REQUIRED WHERE HORIZONTAL LINE POST INSULATORS ARE USED ARE BASED ON BOLT SIZES SHOWN. INSULATOR MAY REQUIRE BOLTS, LOCKNUTS, AND WASHERS OF DIFFERENT SIZES.
- ADDITIONAL POLE BAND IS NOT REQUIRED ON CENTER POLE FOR LINE ANGLES 60°-90°.
- FOR 5°-30° LINE ANGLES ADD TWO ADDITIONAL BANDS (PART 3) OR DEADEND TEES FOR GUYING ON EACH OUTSIDE POLE.
- DEAD END TEES MAY BE USED IN LIEU OF POLE BANDS. CONNECTING LINKS, SHEAVE WHEELS AND GROUNDING CLIP ASSEMBLIES ARE NOT REQUIRED WITH DEAD END TEES. SEE STANDARD DRAWING 41 6138.
- USE X-BRACE EQUAL TO HUGHES BROS NO.2121 WHEN SPECIFIED.

REFERENCE DRAWINGS

WOOD POLE STRUCTURES
GROUNDING DETAILS _____ 41 1012
OVERHEAD GROUND WIRE ASSEMBLY
DETAILS FOR WOOD POLE STRUCTURES _____ 41 1019
TENSION INSULATOR ASSEMBLIES
SINGLE STRING _____ 41 1022
DOUBLE STRING _____ 41 1023
OPTICAL GROUND WIRE ASSEMBLIES _____ 41 1031
TYPE 3TA-1 STRUCTURES
STANDARD GUY ARRANGEMENT _____ 41 6125
TYPICAL DETAILS FOR DEAD END TEES _____ 41 6138

SUPERSEDES DWG NO. 41 6130 A IN PART

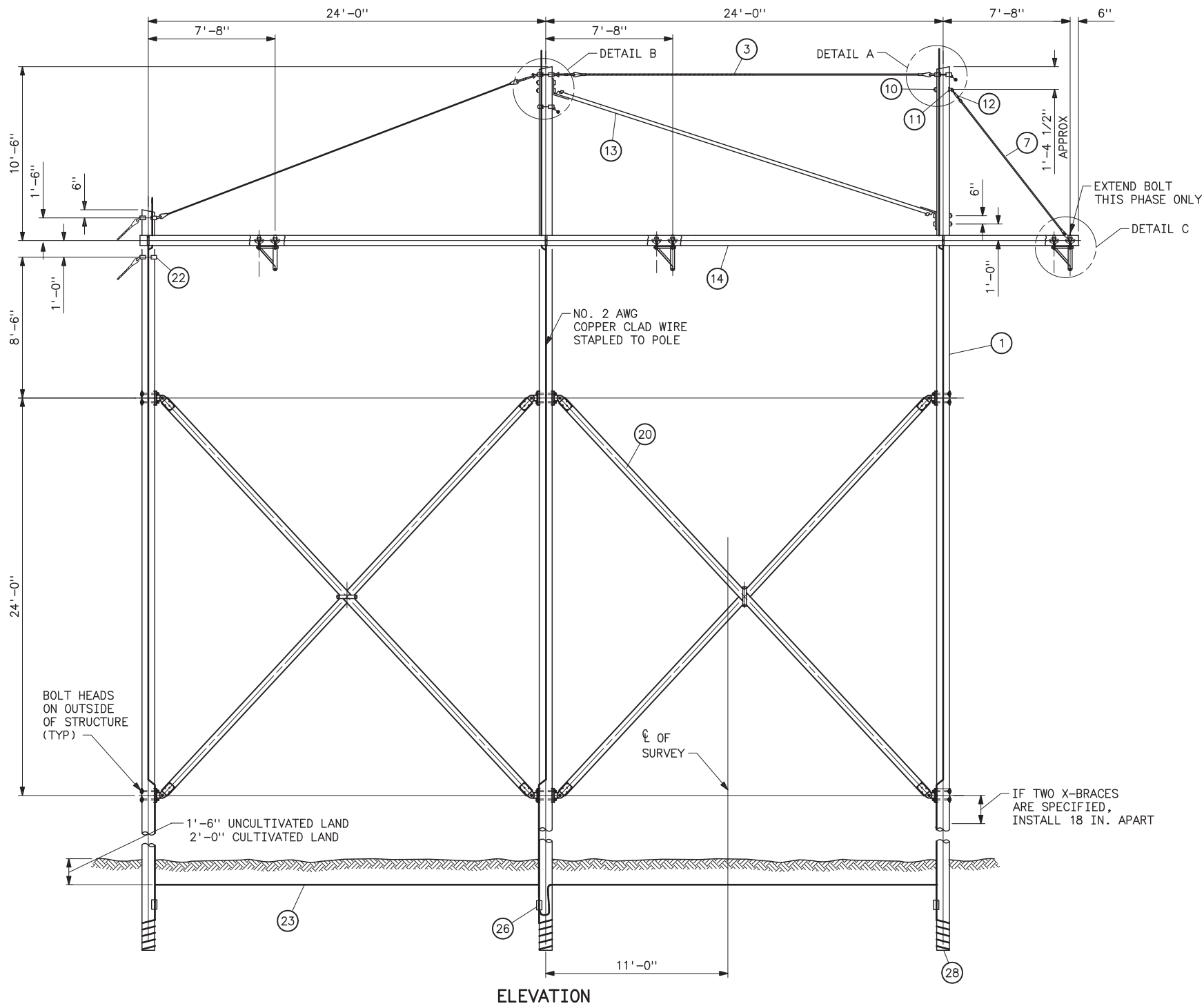
UNITED STATES DEPARTMENT OF ENERGY
WESTERN AREA POWER ADMINISTRATION
CORPORATE SERVICES OFFICE - GOLDEN, COLORADO

STANDARD DESIGNS
TRANSMISSION LINES
TYPE 3TA-1 230KV
3-POLE TENSION STRUCTURE, SHEET 1

DESIGNED KAREN ROWE _____ APPROVED DOUG HANSON _____
CIVIL ENGINEERING
MANAGER

DATE DECEMBER 31, 1999 41 6130-1

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ELEVATION

STRUCTURE MATERIAL

PART NO.	QUANTITY	DESCRIPTION
1	3	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	4	FOUR-WAY STRAIN POLE BAND (EQUAL TO HUGHES BROS. 3103 SERIES)
	8	1/2"x4 LAG SCREWS (2 PER BAND)
	8	1 3/8" FLAT ROUND WASHER WITH 9/16" HOLE (2 PER BAND)
3	AS REQD	7/16" HIGH STRENGTH WIRE (SEE SPECIFICATIONS)
4	4	AUTOMATIC COMPRESSION FITTING WITH SHORT BAIL
5	4	1/4x2x17 CONNECTING LINK PAIR
6	4	2 1/2" SHEAVE WHEEL SUITABLE FOR 7/16" WIRE
7	1	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 25,000 LB MINIMUM ULTIMATE STRENGTH COMPETE W/PINS AND COTTER KEYS
8	1	TWISTED CLIP (EQUAL TO HUGHES BROS. NO. AS2276)
9	1	7/8"x REQD LENGTH BENT BOLT WITH WASHER AND LOCKNUT
10	1	7/8"x14 SHOULDER EYE BOLT WITH LOCK WASHER AND LOCKNUT
11	2	4x4x1/4 CURVED WASHER, 15/16" HOLE
12	3	7/8"x12 CLEVIS-EYE TURNBUCKLE W/BNC (EQUAL TO HUGHES BROS. NO. AS2545-D)
13	1	WOOD VEE BRACE 6 3/8x5 3/8, COMPLETE W/FITTINGS
14	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8x9x56-8
15	6	FLAT GAIN PLATE, 15/16" HOLE
16	6	RIBBED TIE PLATE, 15/16" HOLE
17	3	7/8"x REQD LENGTH DOUBLE ARMING BOLT W/2 LOCK WASHERS & 4 NUTS EACH (USE EXTRA NUTS AS LOCKNUTS)
18	6	ADJUSTABLE SPACER FITTING MADE FROM 5/16" STEEL, COMPLETE W/FITTINGS (HUGHES BROS. 3415.13) OR EQUAL
19	3	SWINGING ANGLE BRACKET, COMPLETE W/FITTINGS (HUGHES BROS. NO. 2821.8, BROOKS MFR NO. 6203 OR EQUAL)
20	2 OR 4	5 1/8x6 GLUED LAMINATED WOOD X-BRACE COMPLETE W/FITTINGS (HUGHES BROS NO. 2121-24-0 OR EQUAL)
21	12	4" SHEAR PLATE FOR 7/8" BOLT
22	1	FOUR-WAY STRAIN POLE BAND (EQUAL TO HUGHES BROS. 3112 SERIES)
30	3	POLE ANTISPLITTING DEVICE EQUAL TO STAR-LOCK MANUFACTURED BY THE BAYNE COMPANY

OVERHEAD GROUND WIRE AND GROUNDING MATERIAL

PART NO.	QUANTITY	DESCRIPTION
23	AS REQD	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2-INCH STAPLES
24	2	OVERHEAD GROUND WIRE SUSPENSION ASSEMBLY
25	2	TWISTED SHACKLE (EQUAL TO HUGHES BROS. NO. 2866)
26	3	PARALLEL GROOVE CLAMP, 2-BOLT COPPER ALLOY FOR NO. 2 AWG COPPER CLAD WIRE
27	5	1/2" GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT). HUGHES 2718.55 OR EQUAL
28	3	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE
29	2	BONDING CLIP (HUGHES BROS. NO. 2727 OF EQUAL)

NOTES

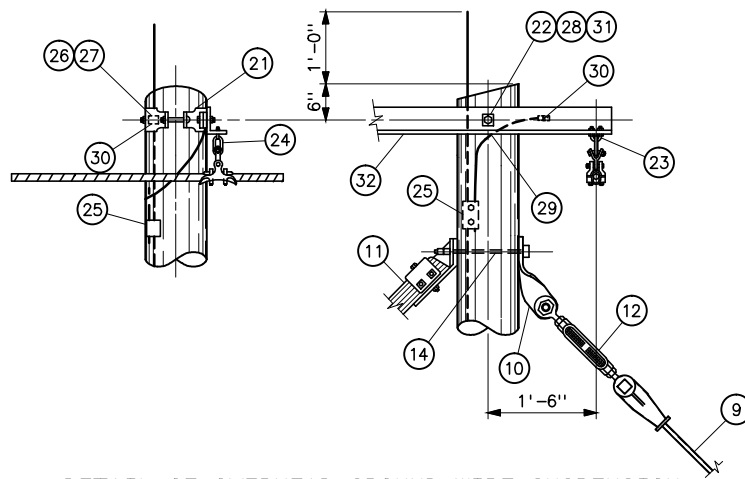
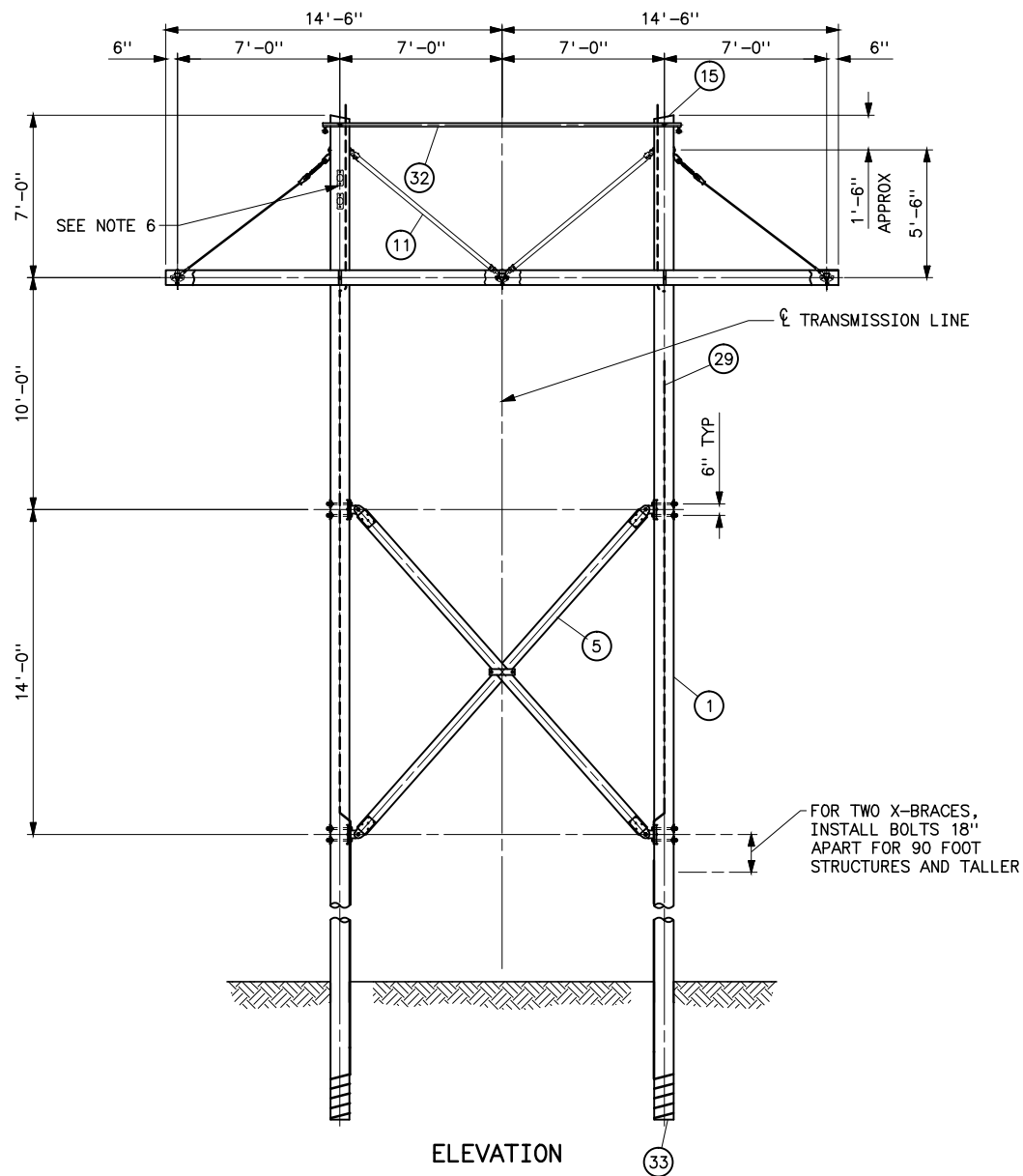
- FITTINGS SHALL INCLUDE BOLTS, NUTS, AND WASHERS.
- ADJUST NUTS ON SWINGING ANGLE BRACKET SO THAT BRACKET IS FREE TO SWING.
- ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
- GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
- HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16 LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
- GUYING REQUIREMENTS ARE SHOWN ON THE PLAN AND PROFILE DRAWINGS.
- HOLES IN CONNECTING LINK PAIRS SHALL BE SIZED 1/16 INCH LARGER THAN CONNECTING BOLTS IN TURNBUCKLES AND STRAIN BANDS. BOLTS SHALL BE FURNISHED WITH 2 NUTS AND COTTER KEY.
- DEAD END TEES MAY BE USED IN LIEU OF POLE BANDS. CONNECTING LINKS, SHEAVE WHEELS AND GROUNDING CLIP ASSEMBLIES ARE NOT REQUIRED WITH DEAD END TEES. SEE STANDARD DRAWING 41 6138.

REFERENCE DRAWINGS

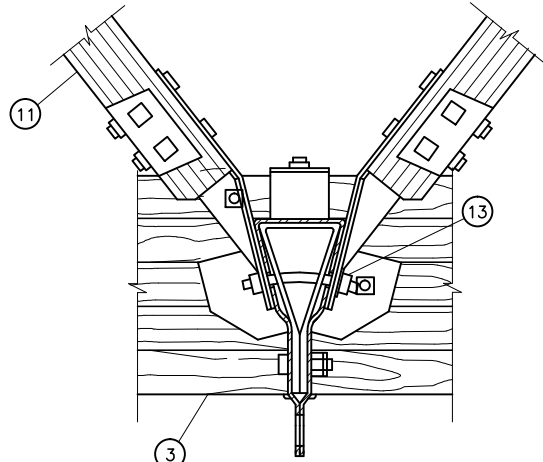
WOOD POLE STRUCTURES	
GROUNDING DETAILS	41 1012
OVERHEAD GROUND WIRE ASSEMBLY	
DETAILS FOR WOOD POLE STRUCTURES	41 1019
SUSPENSION INSULATOR ASSEMBLIES	
SINGLE STRING	41 1020
DOUBLE STRING	41 1021
OPTICAL GROUND WIRE ASSEMBLIES	41 1031
GUY ASSEMBLIES AND DETAILS	41 6047
CROSSARM DRILLING DETAILS	41 6151
GUY ARRANGEMENT FOR TYPE 3AC WITH OGW	41 6054
TYPE 3AC-1 230KV SUSPENSION STRUCTURE	41 6131-2
TYPICAL DETAILS FOR DEAD END TEES	41 6138

A	08-24-06 A7-KKR	CHANGED CROSSARM DEPTH TO 9 INCHES. FIXED DRILLING REFERENCE DRAWING NO.
SUPERSEDES DWG NO. 41 6131A (IN PART)		
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGNS TRANSMISSION LINES TYPE 3AC-1 230KV SUSPENSION STRUCTURE-SHEET 1		
DESIGNED KAREN ROWE APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER		
A	FEBRUARY 29, 2000	41 6131-1

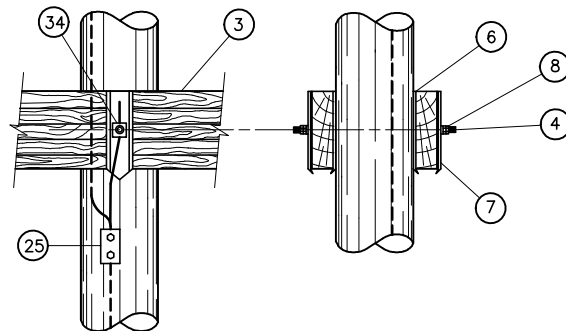
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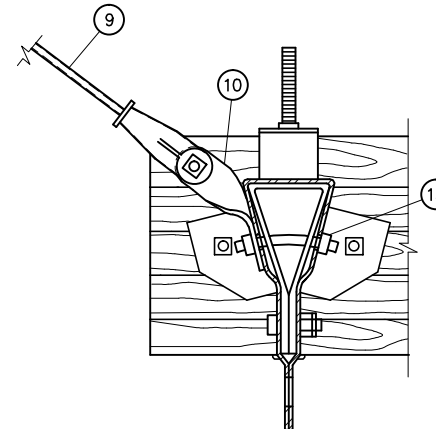
DETAIL OF OVERHEAD GROUND WIRE SUSPENSION ATTACHMENT AND BRACE ATTACHMENT TO POLE



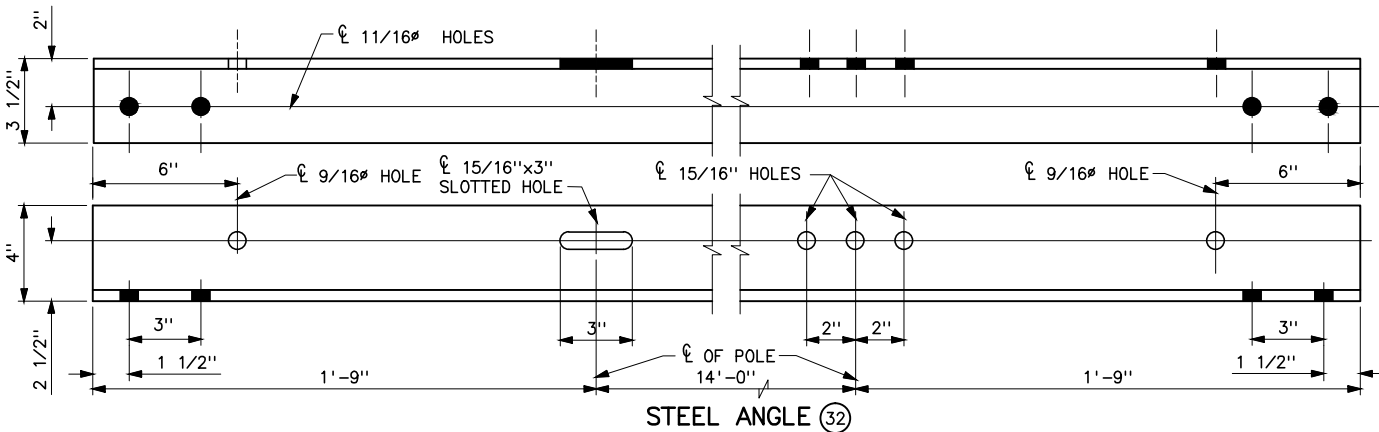
DETAIL OF CENTER PHASE FITTING ATTACHMENTS



DETAIL OF WOOD CROSSARM ATTACHMENT TO POLE



DETAIL OF PHASE FITTING ASSEMBLY



STEEL ANGLE (32)

STRUCTURE MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
1	2	POLE, SEE STRUCTURE LIST FOR HEIGHT
2	3	ADJUSTABLE SPACER FITTINGS COMPLETE WITH BOLTS, NUTS, AND WASHERS (HUGHES BROS. TYPE 3415 OR EQUAL) (SEE NOTE 4)
3	2	GLUED LAMINATED WOOD CROSSARM, 3 1/8x9x29-0
4	2	DOUBLE ARMING BOLT, 7/8x REQUIRED LENGTH WITH 4 NUTS, USE EXTRA NUTS AS LOCKNUTS
5	1	X-BRACE, COMPLETE WITH FITTINGS, 5 1/8x6 (HUGHES BROS. NO. 2121-14-0 OR EQUAL)
6	4	FLAT GAIN PLATE
7	4	RIBBED TIE PLATE
8	4	SPRING LOCK WASHER FOR 7/8 BOLT
9	2	CLEVIS-CLEVIS FIBERGLASS GUY STRAIN, 30,000 LB MINIMUM ULTIMATE STRENGTH COMPLETE WITH BNK
10	4	TWISTED CLIP (HUGHES BROS. NO. AS2276 OF EQUAL)
11	2	WOOD VEE BRACE, 3 1/2x5 1/2, COMPLETE WITH FITTINGS, 5'-6" DROP, SPACING VARIES WITH POLE DIAMETER
12	2	TURNBUCKLE (HUGHES BROS. NO. AS2545-B OF EQUAL)
13	3	7/8x REQUIRED LENGTH BENT STUD WITH LOCK WASHER AND LOCKNUTS, THREADED BOTH SIDES
14	2	7/8x REQUIRED LENGTH WITH LOCK WASHER AND LOCKNUT
15	2	POLE ANTISPLITTING DEVICE EQUAL TO STAR LOCK MANUFACTURED BY THE BAYNE COMPANY

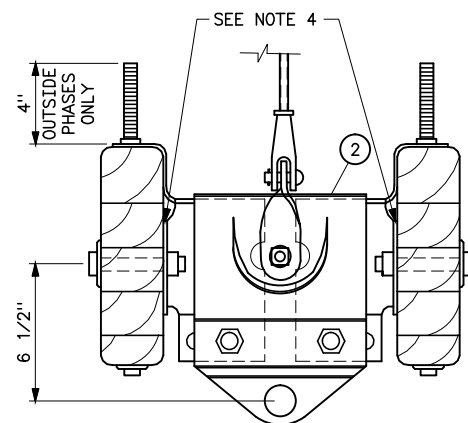
OVERHEAD GROUND WIRE AND GROUNDING MATERIAL		
PART NO.	QUANTITY	DESCRIPTION
21	2	POLE TOP BAND (HUGHES BROS. NO. 2845-B OR EQUAL)
22	2	MACHINE BOLT, 7/8x1-0
23	2	"U" BOLT, 5/8x3 WITH 4 HEX NUTS AND 2 LOCKNUTS
24	2	CHAIN LINK 1/2 (HUGHES BROS. NO. 1329-B1 OR EQUAL)
25	4	PARALLEL GROOVE CLAMP, 2 BOLT COPPER ALLOY OR COPPER CRIMPIT (BURNDY TYPE YC-C OR EQUAL)
26	2	1/2x4 LAG SCREW WITH FETTER DRIVE THREAD (ONE PER BAND)
27	2	1 3/8 FLAT ROUND WASHER WITH 9/16 HOLE (ONE PER BAND)
28	2	LOCKNUT FOR 7/8 BOLT
29	AS REQUIRED	NO. 2 AWG 30% CONDUCTIVITY, COPPER CLAD WIRE ATTACHED WITH 2 INCH STAPLES
30	3	1/2 GROUNDING CLIP ASSEMBLY (INCLUDES BONDING CLIP, CARRIAGE BOLT, NUT AND LOCKNUT)
31	2	SPRING LOCK WASHER FOR 7/8 BOLT
32	1	STEEL ANGLE, 4x3 1/2x3/8x17'-6" LONG
33	1	POLE BOTTOM PLATE (BLACKBURN NO. GP 110 OR EQUAL) ATTACHED TO POLE
34	2	BONDING CLIP (HUGHES BROS. NO. 2727 OR EQUAL)

NOTES

1. STEEL FOR STEEL ANGLES AND FOR BENT PLATES SHALL CONFORM TO ASTM A36. PIECES SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.
2. ALL HOLES IN WOOD SHALL BE BORED 1/8 INCH LARGER THAN THEIR RESPECTIVE BOLT DIAMETERS.
3. GAINING IS NOT REQUIRED IF ADJUSTABLE SPACER FITTING IS USED.
4. HOLE IN PLATE SHALL BE UNTAPPED AND MEASURE 1/16 LARGER THAN BOLT. ADD EXTRA LOCKNUT AND SPRING WASHER FOR EACH CONNECTION.
5. BNK INDICATES BOLT, NUT, AND STAINLESS STEEL COTTER KEY REQUIRED.
6. FOR DEAD ENDING OPGW, SPACE DEAD END TEE TOP BOLT 6" BELOW BRACE ATTACHMENT.

REFERENCE DRAWINGS

WOOD POLE STRUCTURE GROUNDING DETAILS	41 1012
OVERHEAD GROUND WIRE ASSEMBLY DETAILS	
FOR WOOD POLE STRUCTURES	41 1019
SUSPENSION INSULATOR ASSEMBLIES	
SINGLE STRING	41 1020
OPTICAL GROUNDWIRE ASSEMBLIES	41 1031
CROSSARM DRILLING DETAILS HS-1 AND HSB-1 STRUCTURES	41 6113
TYPICAL DETAILS FOR OPTICAL GROUND WIRE SPLICE STRUCTURES	41 6153



A	1-30-12 A7-KKR	CHANGED LENGTH OF STEEL ANGLE.
UNITED STATES DEPARTMENT OF ENERGY WESTERN AREA POWER ADMINISTRATION CORPORATE SERVICES OFFICE - LAKEWOOD, COLORADO		
STANDARD DESIGNS TRANSMISSION LINES TYPE HSB-1 138KV SUSPENSION STRUCTURE		
DESIGNED KAREN ROWE APPROVED DOUG HANSON CIVIL ENGINEERING MANAGER		
A	OCTOBER 26, 2006	41 6159